Set 3D: Diode Small-Signal Analysis

ECEN 325 — Lecture 3 (Diodes) practice set

What this set trains. Circuits where a diode is biased by DC and carries a small AC signal. First find the DC operating point (constant-voltage-drop or exponential model), then replace the diode by its small-signal resistance $r_d = nV_T/I_D$ and solve the AC circuit exactly as in Lectures 1–2; finally add DC and AC. Unless stated otherwise $n = 1.8$ and $V_T = 25\,\text{mV}$. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-3 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1/2 tools):

T1 *Diode models.* Ideal: OFF $\Leftrightarrow V_D < 0$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0$). Constant-voltage-drop (CVD): OFF $\Leftrightarrow V_D < 0.7\,\text{V}$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0.7\,\text{V}$ source). Exponential: $I_D = I_S e^{V_D/nV_T}$, i.e. $V_D = nV_T \ln(I_D/I_S)$, $V_T \approx 25\,\text{mV}$.

T2 *State method.* Assume ON/OFF for every diode, solve the now-linear circuit (KCL, Lecture 2), then check the assumptions; each check is a condition on V_i. The valid states tile the V_i axis.

T3 *Small signal.* Around a DC point I_D the diode is a resistor $r_d = \frac{nV_T}{I_D}$ (valid while $|v_d| \ll nV_T$). Capacitors: open at DC; $\frac{1}{sC}$ in AC; DC supplies are AC ground.

T4 *Rectifier with a filter capacitor* ($f = 60\,\text{Hz}$, $T = \frac{1}{f}$ half-wave, $\frac{1}{2f}$ full-wave): peak $V_p = \hat{V}_S -$ (diode drops in the path); ripple $V_r \approx V_p \frac{T}{RC}$; $I_{D,av} \approx I_L\left(1 + \pi\sqrt{2V_p/V_r}\right)$, $I_{D,max} \approx I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right)$; PIV = the largest reverse voltage a diode sees.

Steps:

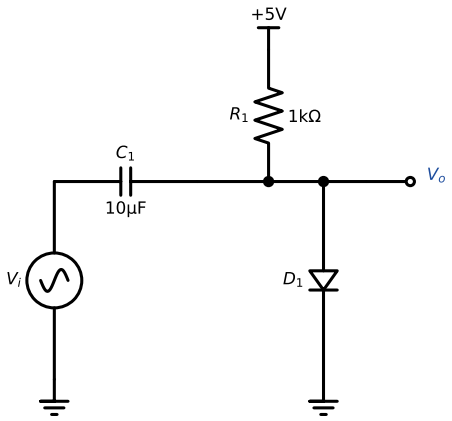
1. *Pick the model* stated in the problem.
2. *Transfer characteristic:* for each ON/OFF state: replace the diodes, solve, check $\Rightarrow$ range of V_i; then join the pieces (they must meet at the breakpoints) and sketch.
3. *Small signal:* DC operating point first, then r_d, then the AC circuit exactly as in Lectures 1–2, then add DC and AC.
4. *Rectifiers:* peak $\rightarrow$ ripple (or C for a given ripple) $\rightarrow$ diode currents $\rightarrow$ PIV $\rightarrow$ transformer ratio.

Problems

Easy problems

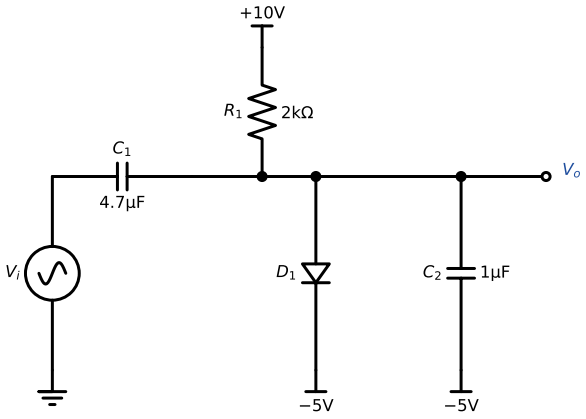
Problem 3D.1 — Easy

In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 5000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



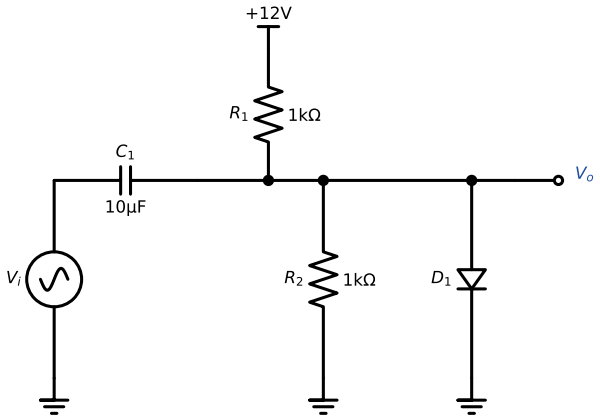
Problem 3D.2 — Easy

In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 10000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



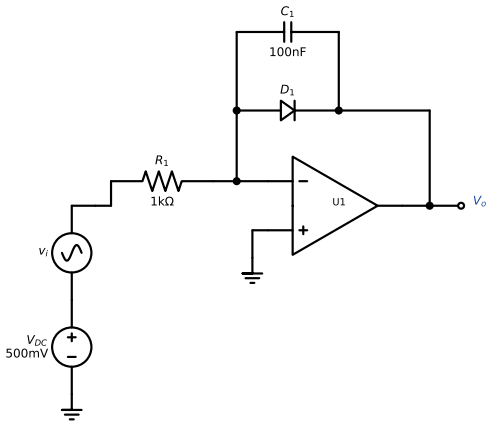
Problem 3D.3 — Easy

In the circuit below, $V_i(t) = 0.001\sin(2\pi \cdot 2000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



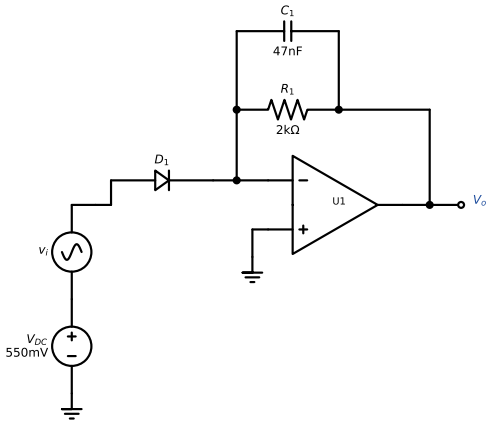
Problem 3D.4 — Easy

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 0.5\,\text{V}$ and $v_i(t) = 0.02\sin(2\pi \cdot 1000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



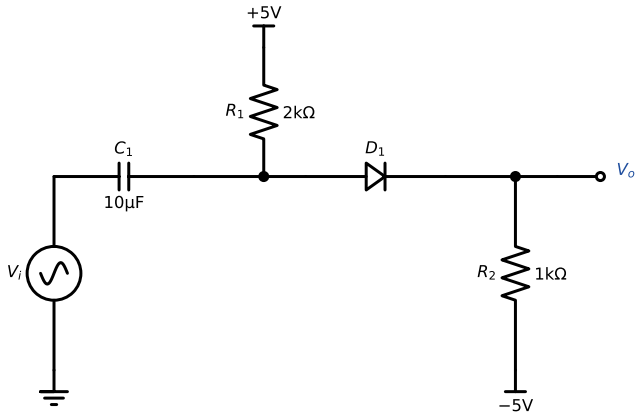
Problem 3D.5 — Easy

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 0.55\,\text{V}$ and $v_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6 \times 10^{-9}\,\text{A}$, $n = 1.8$, $V_T = 25\,\text{mV}$), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Problem 3D.6 — Easy

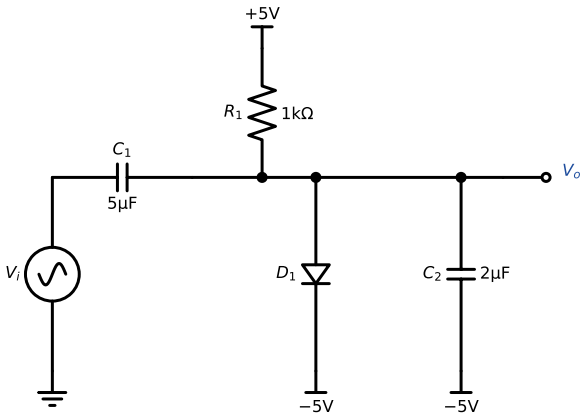
In the circuit below, $V_i(t) = 0.005\sin(2\pi \cdot 1000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Medium problems

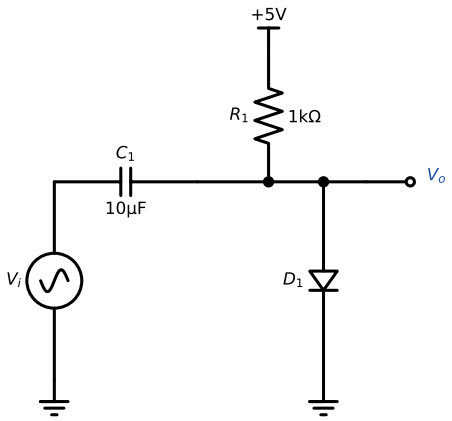
Problem 3D.7 — Medium

In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 10000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



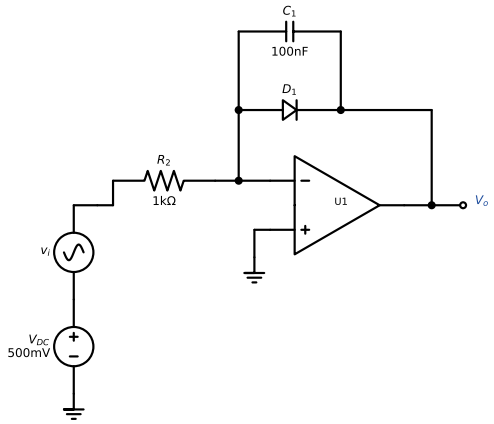
Problem 3D.8 — Medium

In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 5000\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6\times10^{-9}$ A, $n = 1.8$, $V_T = 25\,\text{mV}$), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



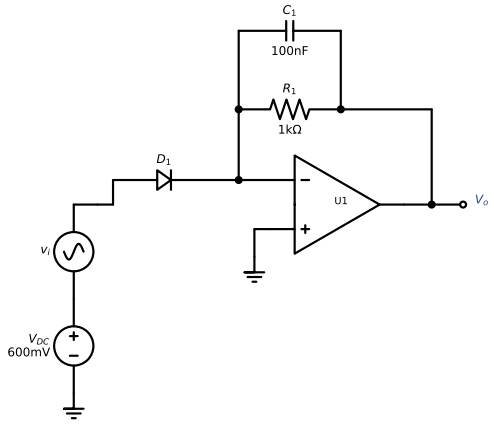
Problem 3D.9 — Medium

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 0.5\,\text{V}$ and $v_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6 \times 10^{-9}\,\text{A}$, $n = 1.8$, $V_T = 25\,\text{mV}$), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



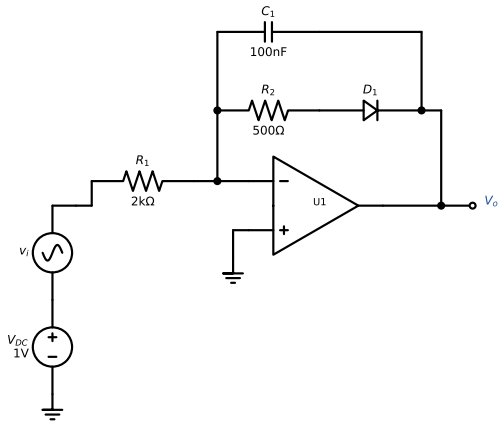
Problem 3D.10 — Medium

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 0.6\,\text{V}$ and $v_i(t) = 0.003\sin(2\pi \cdot 1000\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6 \times 10^{-9}\,\text{A}$, $n = 1.8$, $V_T = 25\,\text{mV}$), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



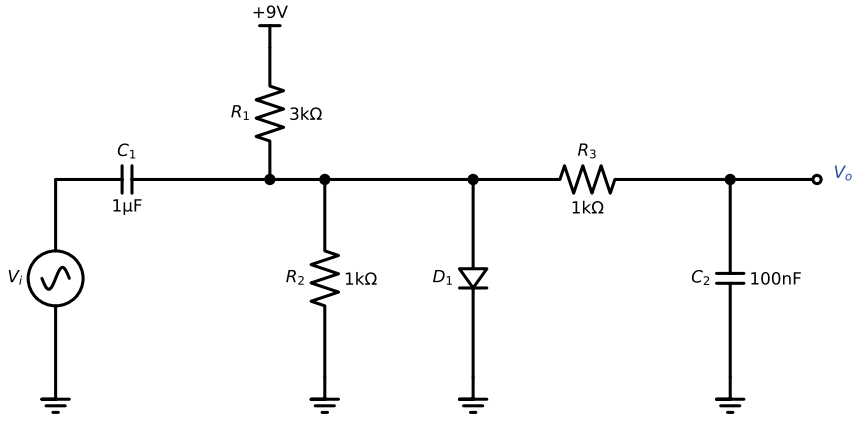
Problem 3D.11 — Medium

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 1\,\text{V}$ and $v_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Problem 3D.12 — Medium

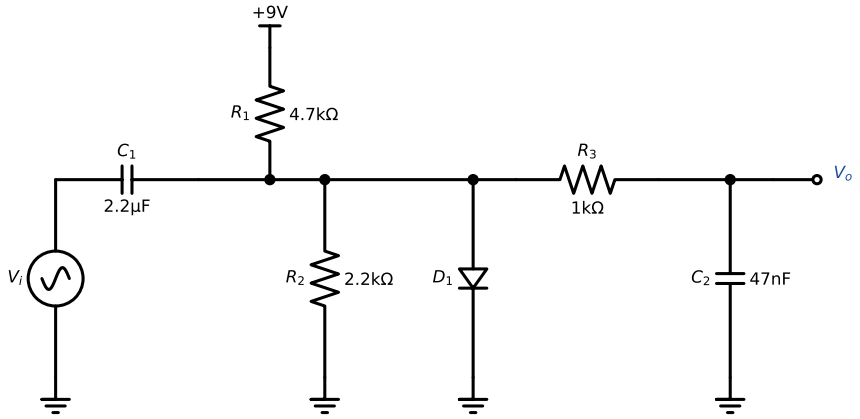
In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Hard problems

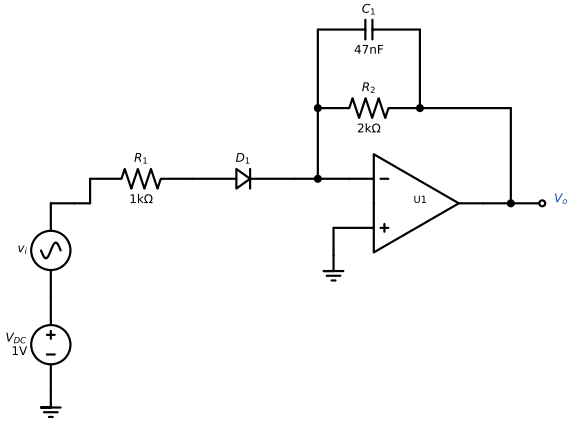
Problem 3D.13 — Hard

In the circuit below, $V_i(t) = 0.003\sin(2\pi \cdot 2000\,t)$. (a) Using the exponential model of the diode ($I_S = 10^{-12}$ A, $n = 1$, $V_T = 25$ mV), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1$, $V_T = 25$ mV for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



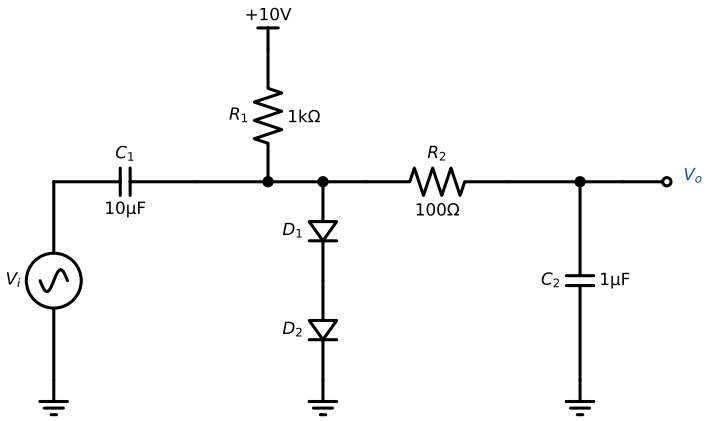
Problem 3D.14 — Hard

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 1$ V and $v_i(t) = 0.01\sin\left(2\pi \cdot 3000\,t + \frac{\pi}{4}\right)$. (a) Using the exponential model of the diode ($I_S = 2.6 \times 10^{-9}$ A, $n = 1.8$, $V_T = 25$ mV), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25$ mV for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



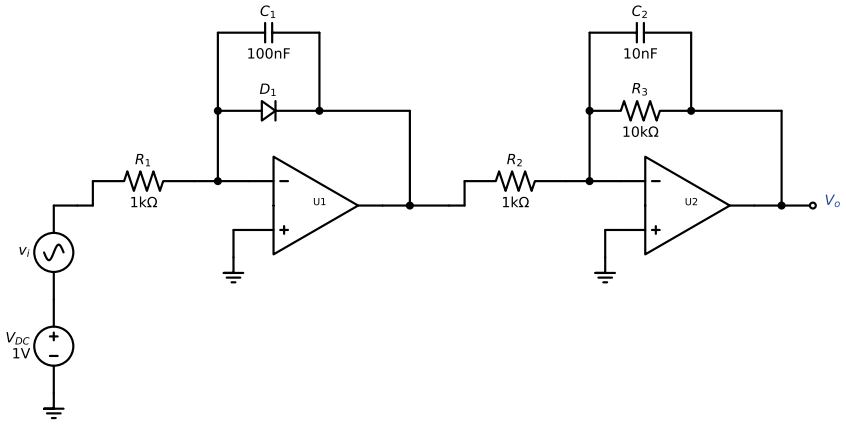
Problem 3D.15 — Hard

In the circuit below, $V_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$. (a) Using the constant-voltage-drop model of the diode (0.7 V), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



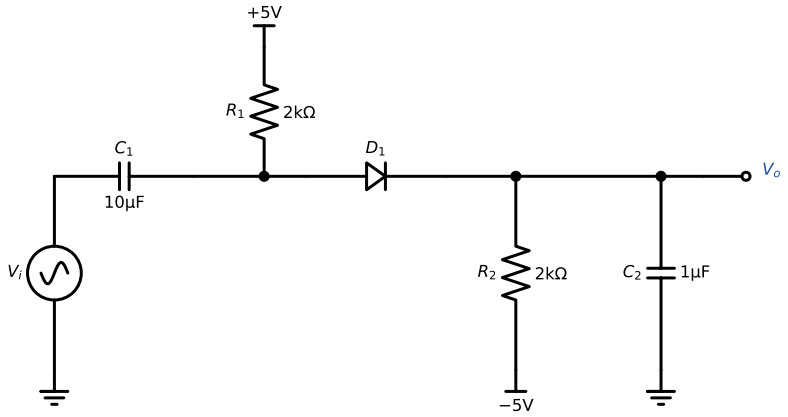
Problem 3D.16 — Hard

In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 1\,\text{V}$ and $v_i(t) = 0.02\sin(2\pi \cdot 2000\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6 \times 10^{-9}\,\text{A}$, $n = 1.8$, $V_T = 25\,\text{mV}$), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25\,\text{mV}$ for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



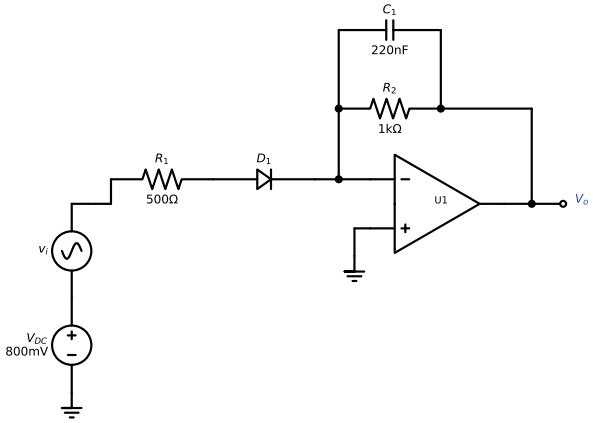
Problem 3D.17 — Hard

In the circuit below, $V_i(t) = 0.005\sin(2\pi \cdot 500\,t)$. (a) Using the exponential model of the diode ($I_S = 2.6\times10^{-9}$ A, $n = 1.8$, $V_T = 25$ mV), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1.8$, $V_T = 25$ mV for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Problem 3D.18 — Hard

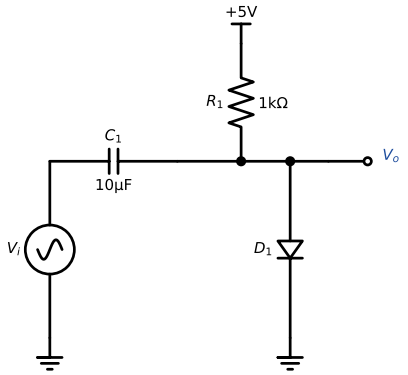
In the circuit below the op-amp is ideal and the input is $V_{DC} + v_i(t)$ with $V_{DC} = 0.8$ V and $v_i(t) = 0.005\cos\left(2\pi \cdot 500\,t - \frac{\pi}{3}\right)$. (a) Using the exponential model of the diode ($I_S = 10^{-12}$ A, $n = 1$, $V_T = 25$ mV), find the DC diode current I_D and the DC output voltage $V_{o,dc}$. (b) Find the small-signal transfer function $\frac{V_o(s)}{V_i(s)}$ (use $n = 1$, $V_T = 25$ mV for r_d) and sketch its magnitude Bode plot. (c) Find the total output $V_o(t)$.



Solutions

Every solution follows the steps of the Lecture-3 pipeline card.

Solution 3D.1 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_o - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - 5}{R_1} + I_{D_1} = 0$$

$$\text{D1 ON}: \qquad V_o - 0 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.7$$
$$I_{D_1} = 4.3 \text{ mA}$$

$I_{D_1} = 4.3\,\text{mA} > 0$ ✓, so the assumption holds.

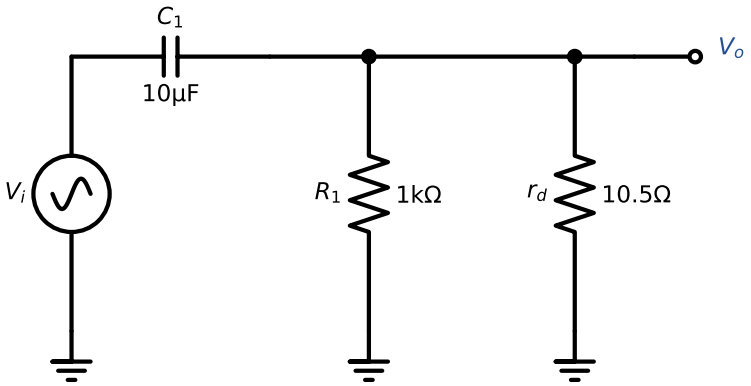
$$\boxed{V_{o,dc} = 700\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{4.3\,\text{mA}} = 10.5\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_1 \parallel r_d = 10.4$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{s}{s + 9660}$$

So:

$$H(s) = \frac{s}{s + 9660}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 9660$; $s + 9660 = 0 \Rightarrow s = -9660$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 9660)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 1.54\text{k}}, \qquad K = 1$$

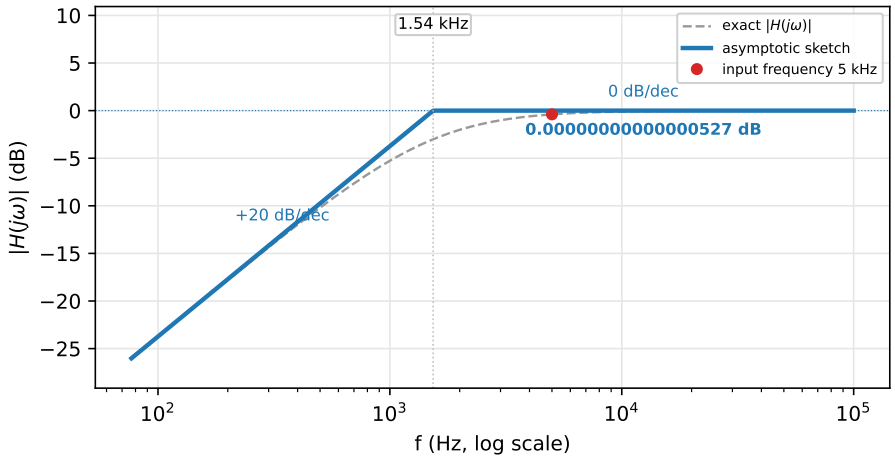
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	9660	1.54 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.04 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.54$ kHz	+20 dB/dec	0 dB at end
$f > 1.54$ kHz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi\cdot 5000\,t)$, so $\omega_{in} = 2\pi\cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 1.54\text{k}}$	0.9559	0.298
total	**0.9559** (product)	**0.298** (sum)

$$|H(j\omega_{in})| = 0.956, \qquad \angle H(j\omega_{in}) = 0.298 \text{ rad}$$
$$v_{o,ac}(t) = 0.956\times 0.002\sin(2\pi\cdot 5000\,t + 0.298) = 0.00191\sin(2\pi\cdot 5000\,t + 0.298)$$

Add the DC part from Step 2 (superposition):

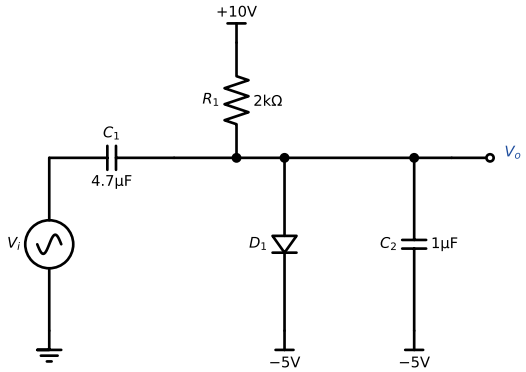
$$\boxed{V_o(t) = 0.7 + 0.00191\sin(2\pi\cdot 5000\,t + 0.298)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.91 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 1.54\text{k}} \quad \text{(corners: 1.54 kHz)}$$
$$V_o(t) = 0.7 + 0.00191\sin(2\pi\cdot 5000\,t + 0.298)\ \text{V}$$

Solution 3D.2 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_o - -5 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - 10}{R_1} + I_{D_1} = 0$$

$$\text{D1 ON}: \qquad V_o - -5 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -4.3$$
$$I_{D_1} = 7.15 \text{ mA}$$

$I_{D_1} = 7.15\,\text{mA} > 0$ ✓, so the assumption holds.

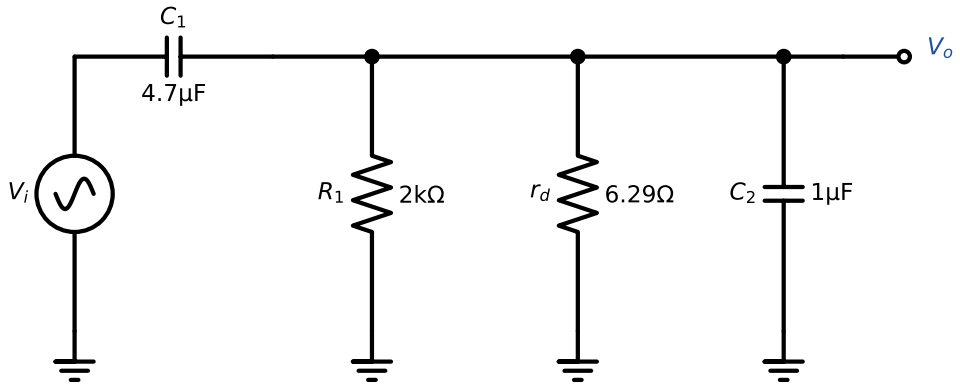
$$\boxed{V_{o,dc} = -4.3\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{7.15\,\text{mA}} = 6.29\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_1 \parallel r_d \parallel \frac{1}{sC_2} = \frac{10^6}{s + 1.59 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{0.825\, s}{s + 28000}$$

So:

$$H(s) = \frac{0.825\, s}{s + 28000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 28000$; $s + 28000 = 0 \Rightarrow s = -28000$.

Factored form (leading coefficient in front):

$$H(s) = 0.825\, \frac{s}{(s + 28000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.825\, \frac{s}{s + 2\pi \cdot 4.45\text{k}}, \qquad K = 0.825$$

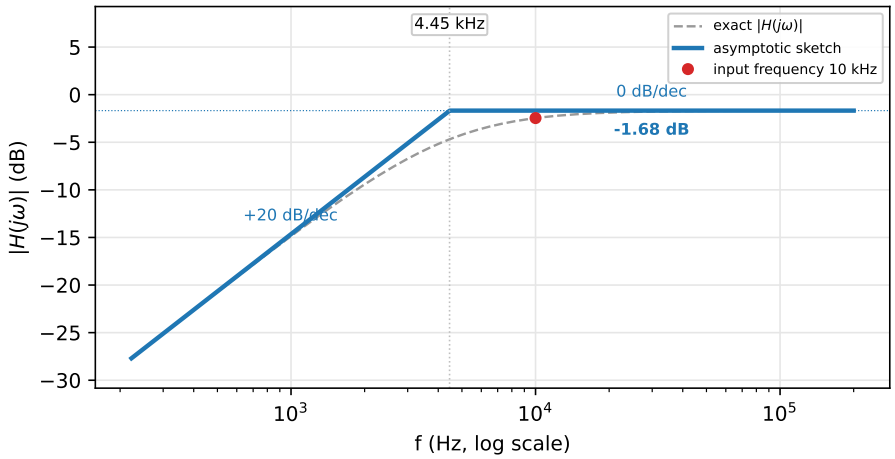
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	28000	4.45 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 2.95 \times 10^{-5}\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 4.45 kHz	+20 dB/dec	-1.68 dB at end
$f >$ 4.45 kHz	0 dB/dec (flat)	-1.68 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.825 = -1.68\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi\cdot 10000\,t)$, so $\omega_{in} = 2\pi\cdot 10000 = 62800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.825	0.8246	0
$\dfrac{s}{s+2\pi\cdot 4.45\text{k}}$	0.9136	0.419
total	**0.7533** (product)	**0.419** (sum)

$$|H(j\omega_{in})| = 0.753, \qquad \angle H(j\omega_{in}) = 0.419 \text{ rad}$$
$$v_{o,ac}(t) = 0.753\times 0.002\sin(2\pi\cdot 10000\,t + 0.419) = 0.00151\sin(2\pi\cdot 10000\,t + 0.419)$$

Add the DC part from Step 2 (superposition):

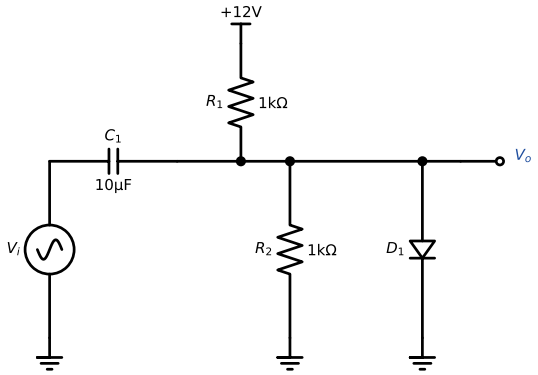
$$\boxed{V_o(t) = -4.3 + 0.00151\sin(2\pi\cdot 10000\,t + 0.419)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.51 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 0.825\,\frac{s}{s+2\pi\cdot 4.45\text{k}} \quad \text{(corners: 4.45 kHz)}$$
$$V_o(t) = -4.3 + 0.00151\sin(2\pi\cdot 10000\,t + 0.419)\ \text{V}$$

Solution 3D.3 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_o - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_o: && \frac{V_o - 12}{R_1} + \frac{V_o}{R_2} + I_{D_1} = 0 \\ &\text{D1 ON}: && V_o - 0 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.7$$
$$I_{D_1} = 10.6 \text{ mA}$$

$I_{D_1} = 10.6\,\text{mA} > 0$ ✓, so the assumption holds.

$$\boxed{V_{o,dc} = 700\,\text{mV}}$$

Step 2 — Small-signal resistance

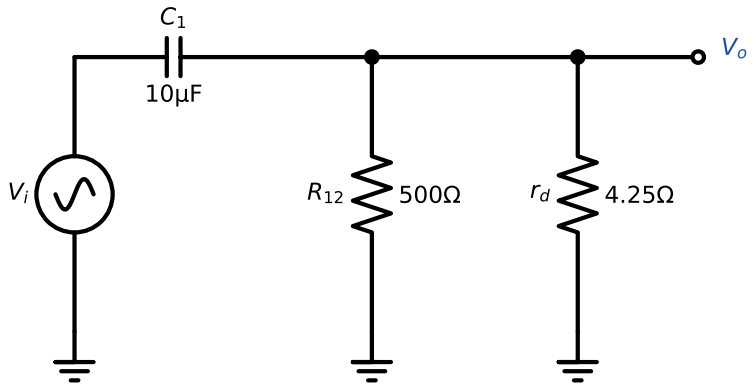
$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{10.6\,\text{mA}} = 4.25\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 1\,\text{k} \parallel 1\,\text{k} = 500\,\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_{12} \parallel r_d = 4.21$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{s}{s + 23800}$$

So:

$$H(s) = \frac{s}{s + 23800}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 23800$; $s + 23800 = 0 \Rightarrow s = -23800$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 23800)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 3.78\text{k}}, \qquad K = 1$$

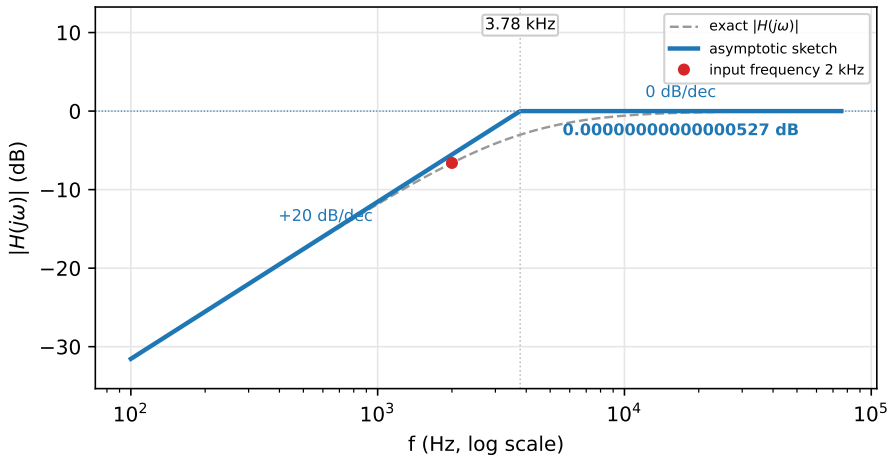
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	23800	3.78 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 4.21 \times 10^{-5}\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.78$ kHz	+20 dB/dec	0 dB at end
$f > 3.78$ kHz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = 0$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.001 \sin(2\pi \cdot 2000\, t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s + 2\pi \cdot 3.78\text{k}}$	0.4676	1.08
total	**0.4676** (product)	**1.08** (sum)

$$|H(j\omega_{in})| = 0.468, \qquad \angle H(j\omega_{in}) = 1.08 \text{ rad}$$

$$v_{o,ac}(t) = 0.468 \times 0.001 \sin(2\pi \cdot 2000\, t + 1.08) = 4.68 \times 10^{-4} \sin(2\pi \cdot 2000\, t + 1.08)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 0.7 + 4.68 \times 10^{-4} \sin(2\pi \cdot 2000\, t + 1.08) \text{ V}}$$

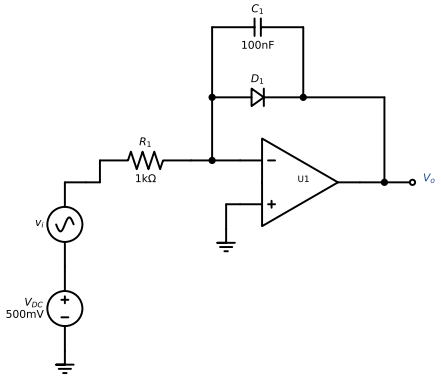
Check the small-signal assumption: the AC voltage across the diode has amplitude about $468\,\mu\text{V}$, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 3.78\text{k}} \quad \text{(corners: 3.78 kHz)}$$

$$V_o(t) = 0.7 + 4.68 \times 10^{-4}\,\sin\left(2\pi \cdot 2000\,t + 1.08\right)\ \text{V}$$

Solution 3D.4 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 500\,\text{mV}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The input current $\frac{V_{DC}-0}{R_1} = 500\,\mu\text{A}$ cannot enter the op-amp: all of it flows through the diode, $I_D = 500\,\mu\text{A}$. Constant-voltage-drop model: $V_D = 0.7\,\text{V}$. The diode current flows from the $-$ input (at 0 V) toward the output, so the output sits V_D below it:

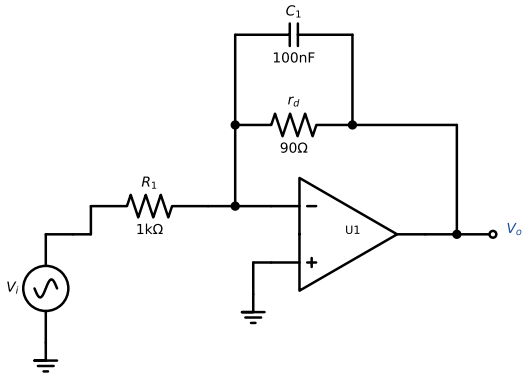
$$V_{o,dc} = -(V_D) = -700\,\text{mV}$$

$$\boxed{V_{o,dc} = -700\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{500\,\mu\text{A}} = 90\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i-0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1}\cdot\frac{V_i}{R_1}$.

$$Z_{f1} = r_d \parallel \frac{1}{sC_1} = \frac{10^7}{s + 1.11\times 10^5}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10000}{s + 1.11 \times 10^5}$$

So:

$$H(s) = \frac{-10000}{s + 1.11 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 1.11 \times 10^5$; $s + 1.11 \times 10^5 = 0 \Rightarrow s = -1.11 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -10000\,\frac{1}{(s + 1.11 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -0.09\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 17.7\text{k}}}, \qquad K = -0.09$$

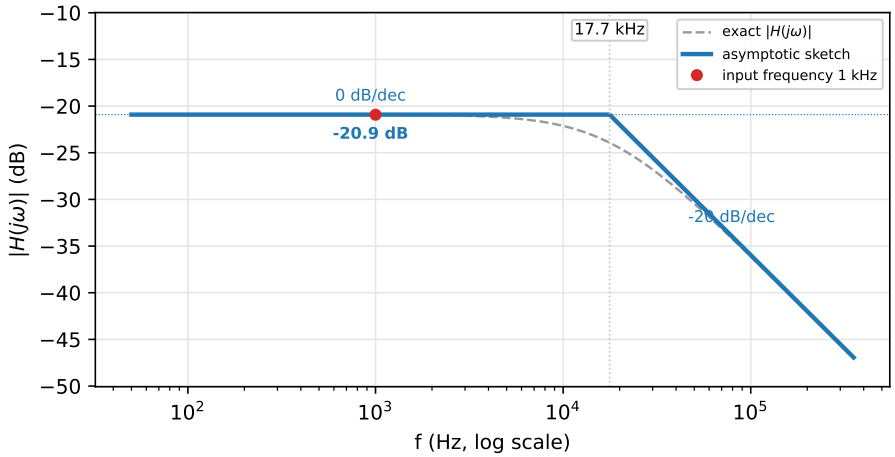
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.11×10^5	17.7 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.09 = -20.9\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 17.7$ kHz	0 dB/dec (flat)	-20.9 dB at end
$f > 17.7$ kHz	-20 dB/dec	-20.9 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.09 = -20.9\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.02\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.09	0.09	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}}$	0.9984	-0.0565
total	**0.08986** (product)	$-$**3.2** (sum)

$$|H(j\omega_{in})| = 0.0899, \qquad \angle H(j\omega_{in}) = -3.2 \text{ rad}$$
$$v_{o,ac}(t) = 0.0899 \times 0.02 \sin(2\pi \cdot 1000\,t - 3.2) = 0.0018 \sin(2\pi \cdot 1000\,t - 3.2)$$

Add the DC part from Step 2 (superposition):

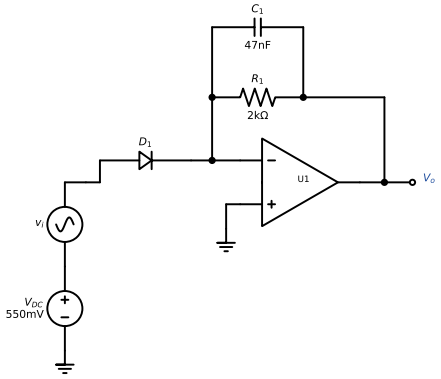
$$\boxed{V_o(t) = -0.7 + 0.0018 \sin(2\pi \cdot 1000\,t - 3.2)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.8 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = -0.09\,\frac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}} \quad \text{(corners: 17.7 kHz)}$$
$$V_o(t) = -0.7 + 0.0018 \sin(2\pi \cdot 1000\,t - 3.2)\ \text{V}$$

Solution 3D.5 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 550\,\text{mV}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The diode sits between V_{DC} and the virtual ground, so $V_D = V_{DC} = 550\,\text{mV}$ and the exponential model (tool T1) gives

$$I_D = I_S\, e^{V_D/nV_T} = 2.6\,\text{nA} \times \exp\left(\frac{0.55}{1.8 \times 0.025}\right) = 528\,\mu\text{A}$$

All of I_D continues through the feedback resistor R_1 (the capacitor is open), so

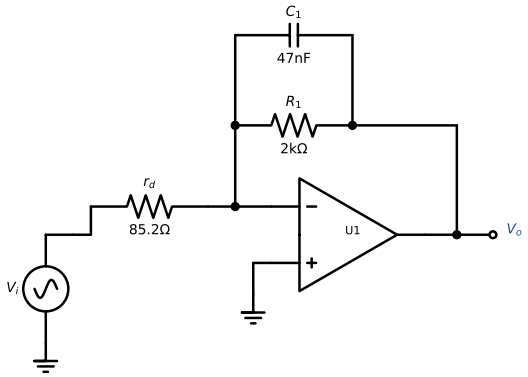
$$V_{o,dc} = 0 - R_1 I_D = -2000 \times 5.285 \times 10^{-4} = -1.06\,\text{V}$$

$$\boxed{V_{o,dc} = -1.06\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{528\,\mu\text{A}} = 85.2\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{r_d}$ through r_d cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{r_d}$.

$$Z_{f1} = R_1 \parallel \frac{1}{sC_1} = \frac{2.13 \times 10^7}{s + 10600}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{r_d} = \frac{-2.5 \times 10^5}{s + 10600}$$

So:

$$H(s) = \frac{-2.5 \times 10^5}{s + 10600}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10600$; $s + 10600 = 0 \Rightarrow s = -10600$.

Factored form (leading coefficient in front):

$$H(s) = -2.5 \times 10^5 \, \frac{1}{(s + 10600)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -23.5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.69\text{k}}}, \qquad K = -23.5$$

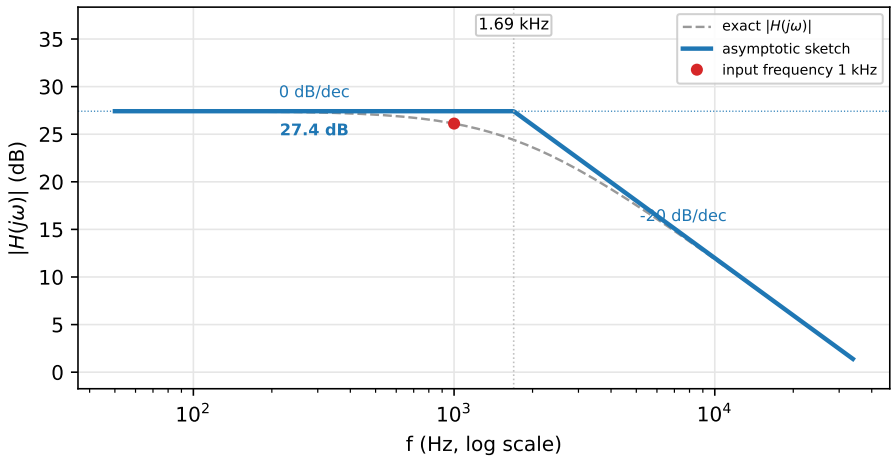
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10600	1.69 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 23.5 = 27.4\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.69$ kHz	0 dB/dec (flat)	27.4 dB at end
$f > 1.69$ kHz	−20 dB/dec	27.4 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 23.5 = 27.4\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-23.5	23.49	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.69\text{k}}}$	0.861	-0.533
total	**20.22** (product)	$-$**3.68** (sum)

$$|H(j\omega_{in})| = 20.2, \qquad \angle H(j\omega_{in}) = -3.68 \text{ rad}$$
$$v_{o,ac}(t) = 20.2 \times 0.002 \sin(2\pi \cdot 1000\,t - 3.68) = 0.0404 \sin(2\pi \cdot 1000\,t - 3.68)$$

Add the DC part from Step 2 (superposition):

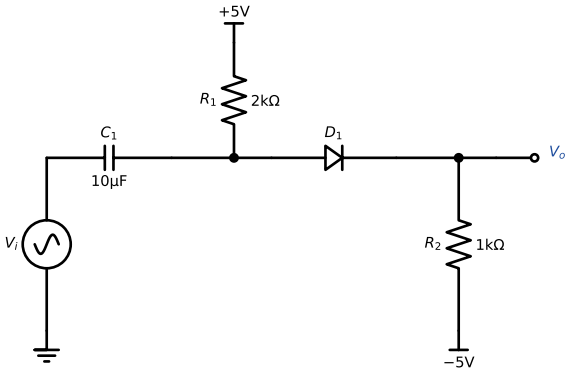
$$\boxed{V_o(t) = -1.06 + 0.0404 \sin(2\pi \cdot 1000\,t - 3.68) \text{ V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 2 mV, well below $nV_T = 45$ mV ✓.

Answer

$$H(s) = -23.5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.69\text{k}}} \quad \text{(corners: 1.69 kHz)}$$
$$V_o(t) = -1.06 + 0.0404 \sin(2\pi \cdot 1000\,t - 3.68) \text{ V}$$

Solution 3D.6 **Easy**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_1 - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_1: && \frac{V_1 - 5}{R_1} + I_{D_1} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - -5}{R_2} - I_{D_1} = 0 \\ &\text{D1 ON}: && V_1 - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -1.9 \\ I_{D_1} &= 3.1 \text{ mA} \end{aligned}$$

$I_{D_1} = 3.1\,\text{mA} > 0$ ✓, so the assumption holds.

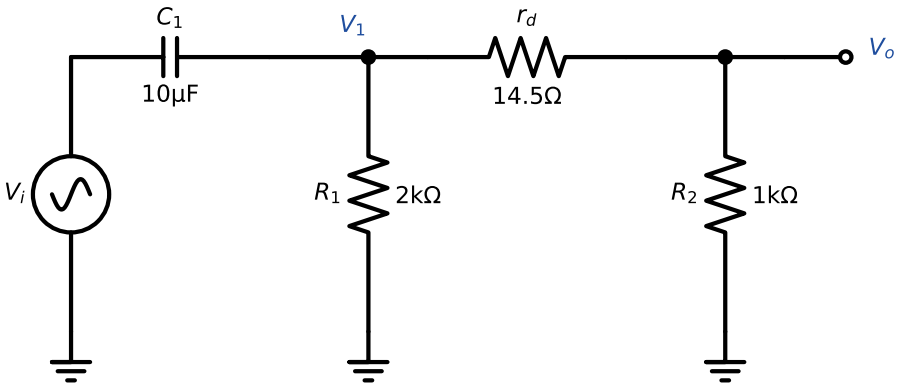
$$\boxed{V_{o,dc} = -1.9\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{3.1\,\text{mA}} = 14.5\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel (r_d + R_2) = 673$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s}{s + 149}$$
$$\frac{V_o}{V_1} = \frac{R_2}{r_d + R_2} = 0.986$$

Multiply the links and simplify:

$$H(s) = \frac{0.986\, s}{s + 149}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 149$; $s + 149 = 0 \Rightarrow s = -149$.

Factored form (leading coefficient in front):

$$H(s) = 0.986\, \frac{s}{(s + 149)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.986\, \frac{s}{s + 2\pi \cdot 23.6}, \qquad K = 0.986$$

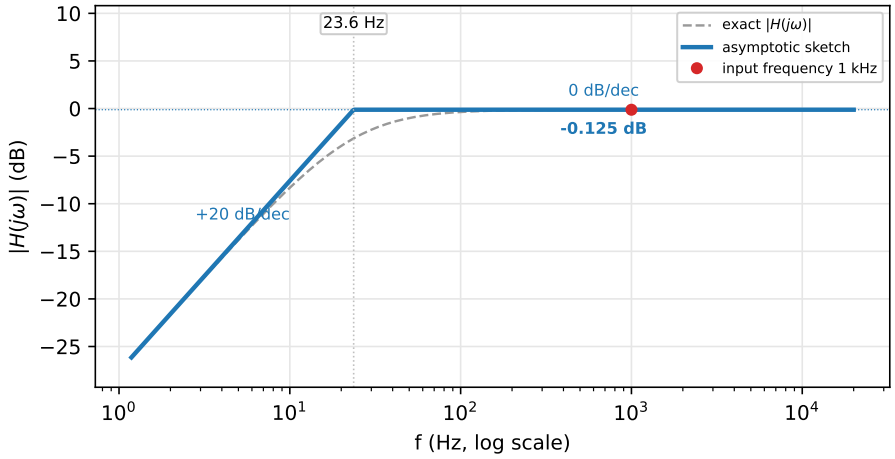
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	149	23.6 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00663\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 23.6$ Hz	+20 dB/dec	-0.125 dB at end
$f > 23.6$ Hz	0 dB/dec (flat)	-0.125 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.986 = -0.125$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.005\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.986	0.9857	0
$\dfrac{s}{s + 2\pi \cdot 23.6}$	0.9997	0.0236
total	**0.9854** (product)	**0.0236** (sum)

$$|H(j\omega_{in})| = 0.985, \qquad \angle H(j\omega_{in}) = 0.0236 \text{ rad}$$

$$v_{o,ac}(t) = 0.985 \times 0.005 \sin(2\pi \cdot 1000\,t + 0.0236) = 0.00493 \sin(2\pi \cdot 1000\,t + 0.0236)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = -1.9 + 0.00493 \sin(2\pi \cdot 1000\,t + 0.0236)\ \text{V}}$$

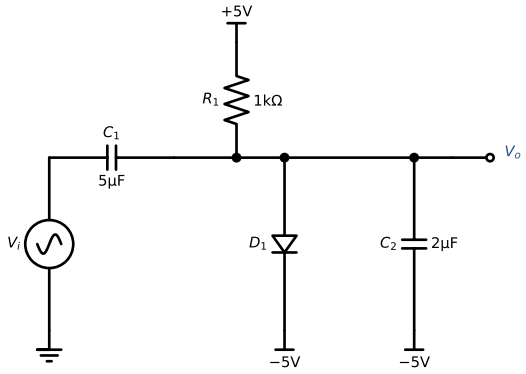
Check the small-signal assumption: the AC voltage across the diode has amplitude about $71.5\,\mu$V, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$H(s) = 0.986 \dfrac{s}{s + 2\pi \cdot 23.6}$ (corners: 23.6 Hz)

$V_o(t) = -1.9 + 0.00493 \sin(2\pi \cdot 1000\,t + 0.0236)$ V

Solution 3D.7 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_o - -5 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - 5}{R_1} + I_{D_1} = 0$$

$$\text{D1 ON}: \qquad V_o - -5 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -4.3$$
$$I_{D_1} = 9.3 \text{ mA}$$

$I_{D_1} = 9.3\,\text{mA} > 0$ ✓, so the assumption holds.

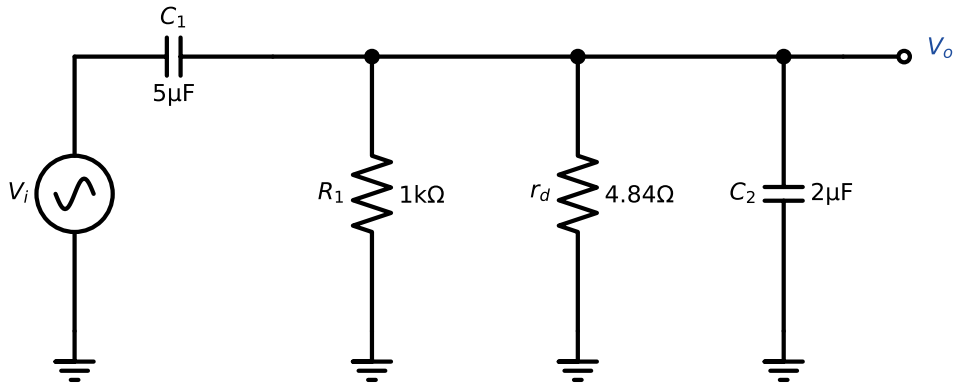
$$\boxed{V_{o,dc} = -4.3\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{9.3\,\text{mA}} = 4.84\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_1 \parallel r_d \parallel \frac{1}{sC_2} = \frac{5 \times 10^5}{s + 1.04 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{0.714\,s}{s + 29700}$$

So:

$$H(s) = \frac{0.714\,s}{s + 29700}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 29700$; $s + 29700 = 0 \Rightarrow s = -29700$.

Factored form (leading coefficient in front):

$$H(s) = 0.714\,\frac{s}{(s + 29700)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.714\,\frac{s}{s + 2\pi \cdot 4.72\text{k}}, \qquad K = 0.714$$

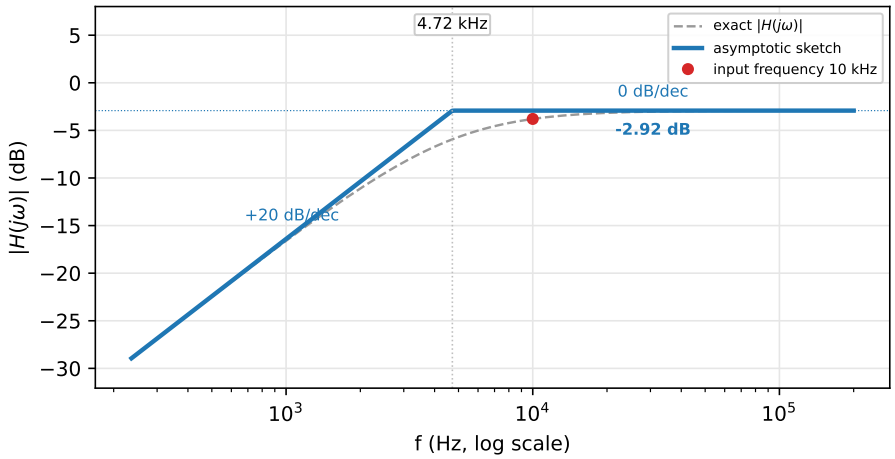
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	29700	4.72 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 2.41 \times 10^{-5}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 4.72 kHz	+20 dB/dec	-2.92 dB at end
$f >$ 4.72 kHz	0 dB/dec (flat)	-2.92 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.714 = -2.92\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi\cdot 10000\,t)$, so $\omega_{in} = 2\pi\cdot 10000 = 62800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.714	0.7143	0
$\dfrac{s}{s+2\pi\cdot 4.72\text{k}}$	0.9043	0.441
total	**0.6459** (product)	**0.441** (sum)

$$|H(j\omega_{in})| = 0.646, \qquad \angle H(j\omega_{in}) = 0.441 \text{ rad}$$
$$v_{o,ac}(t) = 0.646\times 0.002\sin(2\pi\cdot 10000\,t + 0.441) = 0.00129\sin(2\pi\cdot 10000\,t + 0.441)$$

Add the DC part from Step 2 (superposition):

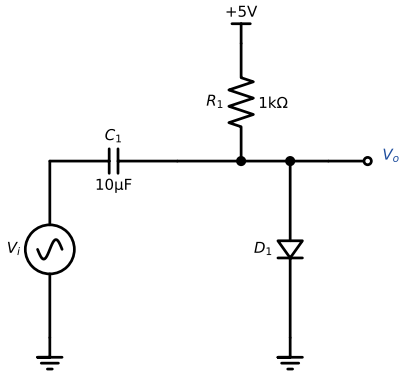
$$\boxed{V_o(t) = -4.3 + 0.00129\sin(2\pi\cdot 10000\,t + 0.441)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.29 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 0.714\,\frac{s}{s+2\pi\cdot 4.72\text{k}} \quad \text{(corners: 4.72 kHz)}$$
$$V_o(t) = -4.3 + 0.00129\sin(2\pi\cdot 10000\,t + 0.441)\ \text{V}$$

Solution 3D.8 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1). With the exponential model the diode voltage is unknown, so first reduce the rest of the circuit to what the diode sees: a Thevenin source. Remove the diode and solve the circuit (Lecture 2 KCL): the open-circuit voltage across it is $V_{th} = 5\,\text{V}$. Short the diode instead: the current is $I_{sc} = 5\,\text{mA}$. So $R_{th} = \frac{V_{th}}{I_{sc}} = 1\,\text{k}\Omega$.

Two equations, two unknowns: the circuit gives $I_D = \frac{V_{th} - V_D}{R_{th}}$ and the diode gives $V_D = nV_T \ln \frac{I_D}{I_S}$ (tool T1, $n = 1.8$, $V_T = 25\,\text{mV}$, $I_S = 2.6\,\text{nA}$). Solve by iteration: guess $V_D = 0.7\,\text{V}$, compute I_D, update V_D, repeat until nothing changes:

step	V_D used (V)	circuit $\Rightarrow I_D$	diode $\Rightarrow$ new V_D (V)
1	0.7	$I_D = \frac{5-0.7}{1000} = 4.3\,\text{mA}$	0.6443
2	0.6443	$I_D = \frac{5-0.6443}{1000} = 4.36\,\text{mA}$	0.6449
3	0.6449	$I_D = \frac{5-0.6449}{1000} = 4.36\,\text{mA}$	0.6449
4	0.6449	$I_D = \frac{5-0.6449}{1000} = 4.36\,\text{mA}$	0.6449
5	0.6449	$I_D = \frac{5-0.6449}{1000} = 4.36\,\text{mA}$	0.6449

Converged: $V_D = 0.6449\,\text{V}$, $I_D = 4.36\,\text{mA}$. With this V_D the rest of the DC circuit gives $V_{o,dc} = 645\,\text{mV}$.

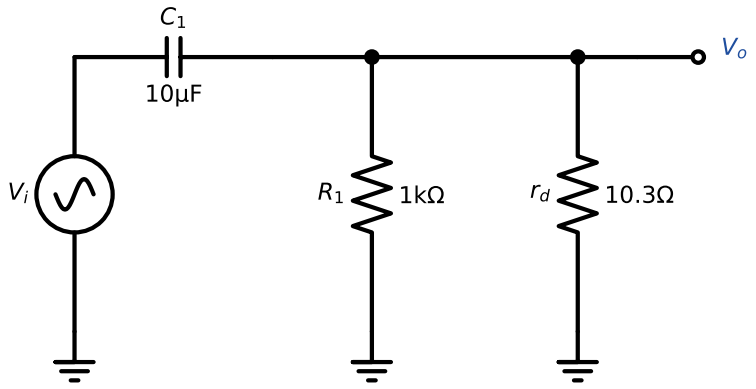
$$\boxed{V_{o,dc} = 645\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{4.36\,\text{mA}} = 10.3\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_1 \parallel r_d = 10.2$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{s}{s + 9780}$$

So:

$$H(s) = \frac{s}{s + 9780}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 9780$; $s + 9780 = 0 \Rightarrow s = -9780$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 9780)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 1.56\text{k}}, \qquad K = 1$$

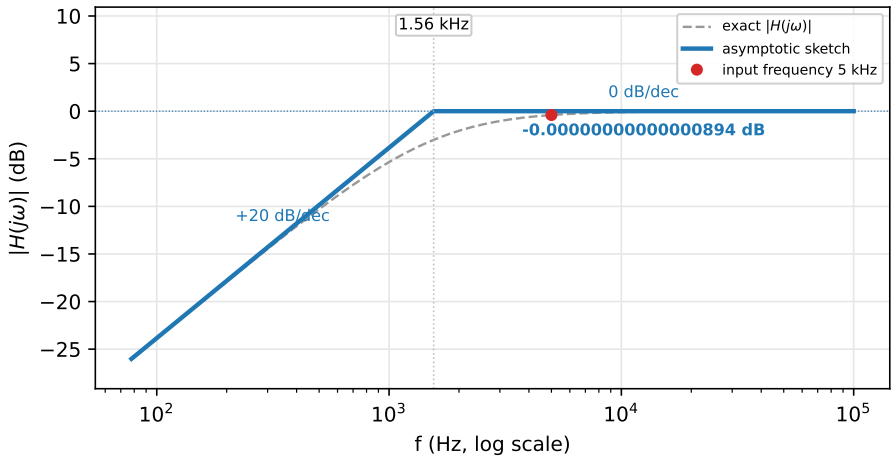
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	9780	1.56 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.02 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.56$ kHz	+20 dB/dec	0 dB at end
$f > 1.56$ kHz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s + 2\pi \cdot 1.56\text{k}}$	0.9548	0.302
total	**0.9548** (product)	**0.302** (sum)

$$|H(j\omega_{in})| = 0.955, \qquad \angle H(j\omega_{in}) = 0.302 \text{ rad}$$
$$v_{o,ac}(t) = 0.955 \times 0.002 \sin(2\pi \cdot 5000\,t + 0.302) = 0.00191 \sin(2\pi \cdot 5000\,t + 0.302)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 0.645 + 0.00191 \sin(2\pi \cdot 5000\,t + 0.302) \text{ V}}$$

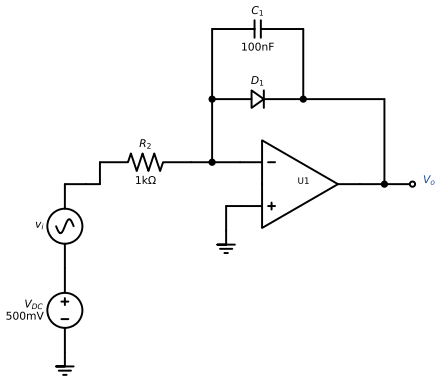
Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.91 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 1.56\text{k}} \quad \text{(corners: 1.56 kHz)}$$

$$V_o(t) = 0.645 + 0.00191\,\sin\left(2\pi \cdot 5000\,t + 0.302\right)\ \text{V}$$

Solution 3D.9 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 500\,\text{mV}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The input current $\frac{V_{DC}-0}{R_2} = 500\,\mu\text{A}$ cannot enter the op-amp: all of it flows through the diode, $I_D = 500\,\mu\text{A}$. Exponential model (tool T1):

$$V_D = nV_T \ln \frac{I_D}{I_S} = 1.8 \times 0.025 \times \ln \frac{5 \times 10^{-4}}{2.6 \times 10^{-9}} = 548\,\text{mV}$$

The diode current flows from the $-$ input (at 0 V) toward the output, so the output sits V_D below it:

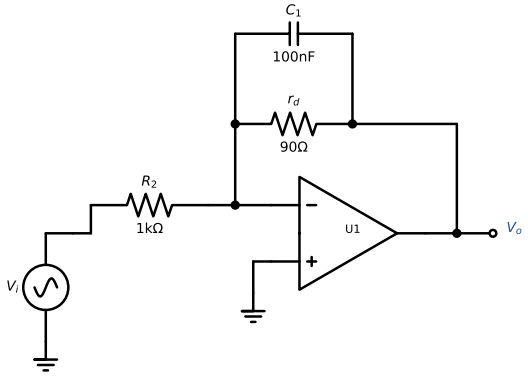
$$V_{o,dc} = -(V_D) = -548\,\text{mV}$$

$$\boxed{V_{o,dc} = -548\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{500\,\mu\text{A}} = 90\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i-0}{R_2}$ through R_2 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{R_2}$.

$$Z_{f1} = r_d \parallel \frac{1}{sC_1} = \frac{10^7}{s + 1.11 \times 10^5}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_2} = \frac{-10000}{s + 1.11 \times 10^5}$$

So:

$$H(s) = \frac{-10000}{s + 1.11 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 1.11 \times 10^5$; $s + 1.11 \times 10^5 = 0 \Rightarrow s = -1.11 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -10000\,\frac{1}{(s + 1.11 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -0.09\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 17.7\text{k}}}, \qquad K = -0.09$$

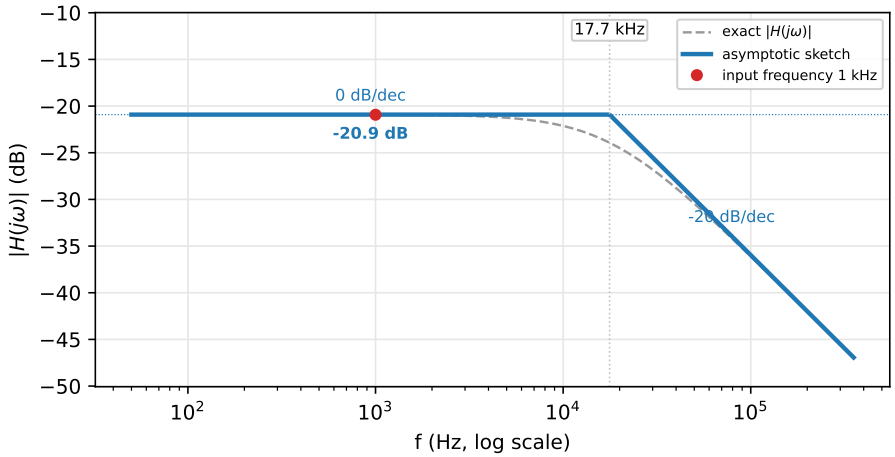
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.11×10^5	17.7 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.09 = -20.9\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 17.7$ kHz	0 dB/dec (flat)	-20.9 dB at end
$f > 17.7$ kHz	-20 dB/dec	-20.9 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.09 = -20.9\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.09	0.09	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}}$	0.9984	-0.0565
total	**0.08986** (product)	$-$**3.2** (sum)

$$|H(j\omega_{in})| = 0.0899, \qquad \angle H(j\omega_{in}) = -3.2 \text{ rad}$$
$$v_{o,ac}(t) = 0.0899 \times 0.05\,\sin(2\pi \cdot 1000\,t - 3.2) = 0.00449\,\sin(2\pi \cdot 1000\,t - 3.2)$$

Add the DC part from Step 2 (superposition):

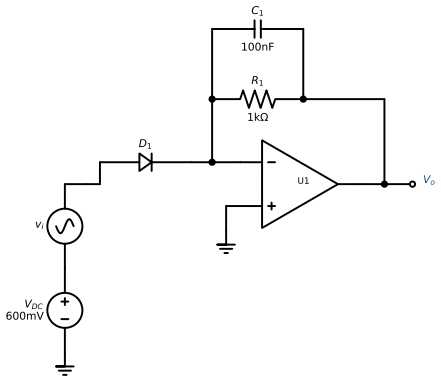
$$\boxed{V_o(t) = -0.548 + 0.00449\,\sin(2\pi \cdot 1000\,t - 3.2)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 4.49 mV, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = -0.09\,\frac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}} \quad \text{(corners: 17.7 kHz)}$$
$$V_o(t) = -0.548 + 0.00449\,\sin(2\pi \cdot 1000\,t - 3.2)\ \text{V}$$

Solution 3D.10 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 600\,\text{mV}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The diode sits between V_{DC} and the virtual ground, so $V_D = V_{DC} = 600\,\text{mV}$ and the exponential model (tool T1) gives

$$I_D = I_S\, e^{V_D/nV_T} = 2.6\,\text{nA} \times \exp\left(\frac{0.6}{1.8 \times 0.025}\right) = 1.61\,\text{mA}$$

All of I_D continues through the feedback resistor R_1 (the capacitor is open), so

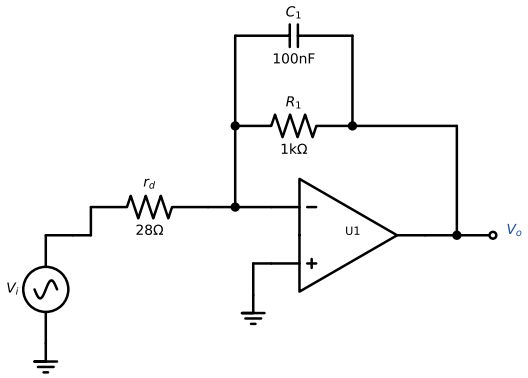
$$V_{o,dc} = 0 - R_1 I_D = -1000 \times 0.001605 = -1.61\,\text{V}$$

$$\boxed{V_{o,dc} = -1.61\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{1.61\,\text{mA}} = 28\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{r_d}$ through r_d cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{r_d}$.

$$Z_{f1} = R_1 \parallel \frac{1}{sC_1} = \frac{10^7}{s + 10000}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{r_d} = \frac{-3.57 \times 10^5}{s + 10000}$$

So:

$$H(s) = \frac{-3.57 \times 10^5}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = -3.57 \times 10^5 \, \frac{1}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -35.7 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = -35.7$$

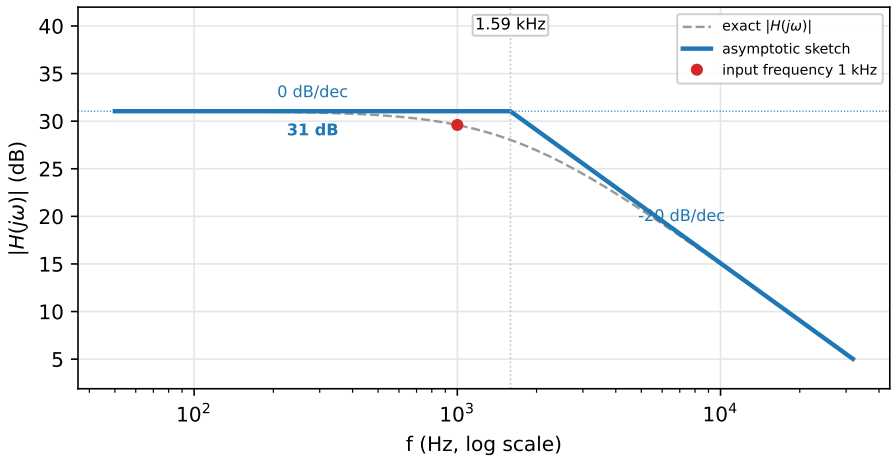
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 35.7 = 31$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	31 dB at end
$f > 1.59$ kHz	−20 dB/dec	31 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 35.7 = 31$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.003\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-35.7	35.67	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.8467	-0.561
total	**30.21** (product)	$-$**3.7** (sum)

$$|H(j\omega_{in})| = 30.2, \qquad \angle H(j\omega_{in}) = -3.7 \text{ rad}$$
$$v_{o,ac}(t) = 30.2 \times 0.003 \sin(2\pi \cdot 1000\,t - 3.7) = 0.0906 \sin(2\pi \cdot 1000\,t - 3.7)$$

Add the DC part from Step 2 (superposition):

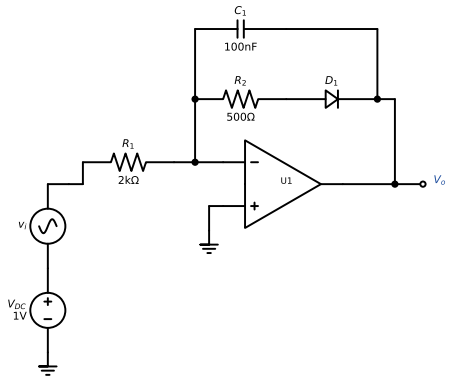
$$\boxed{V_o(t) = -1.61 + 0.0906 \sin(2\pi \cdot 1000\,t - 3.7) \text{ V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 3 mV, well below $nV_T = 45$ mV ✓.

Answer

$$H(s) = -35.7\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}} \quad \text{(corners: 1.59 kHz)}$$
$$V_o(t) = -1.61 + 0.0906 \sin(2\pi \cdot 1000\,t - 3.7) \text{ V}$$

Solution 3D.11 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 1\,\text{V}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The input current $\frac{V_{DC}-0}{R_1} = 500\,\mu\text{A}$ cannot enter the op-amp: all of it flows through the diode, $I_D = 500\,\mu\text{A}$. Constant-voltage-drop model: $V_D = 0.7\,\text{V}$. The diode current flows from the $-$ input (at 0 V) toward the output, so the output sits $V_D + R_2 I_D$ below it:

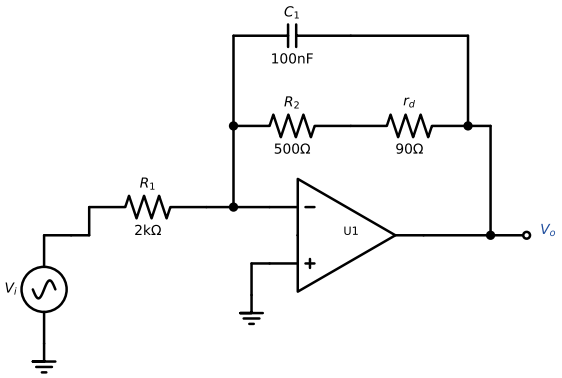
$$V_{o,dc} = -(V_D + R_2 I_D) = -950\,\text{mV}$$

$$\boxed{V_{o,dc} = -950\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{500\,\mu\text{A}} = 90\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i-0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1}\cdot\frac{V_i}{R_1}$.

$$Z_{f1} = (R_2 + r_d) \parallel \frac{1}{sC_1} = \frac{10^7}{s + 16900}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-5000}{s + 16900}$$

So:

$$H(s) = \frac{-5000}{s + 16900}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 16900$; $s + 16900 = 0 \Rightarrow s = -16900$.

Factored form (leading coefficient in front):

$$H(s) = -5000\,\frac{1}{(s + 16900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -0.295\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 2.7\text{k}}}, \qquad K = -0.295$$

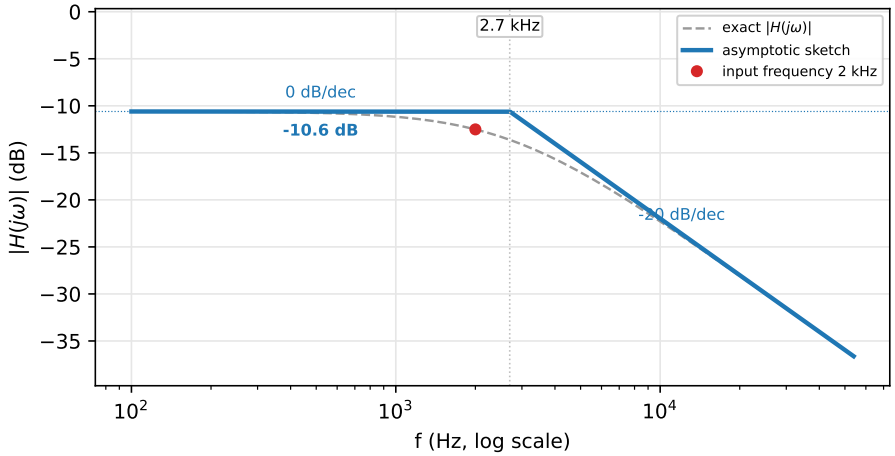
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	16900	2.7 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.295 = -10.6\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 2.7$ kHz	0 dB/dec (flat)	-10.6 dB at end
$f > 2.7$ kHz	−20 dB/dec	-10.6 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.295 = -10.6\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.295	0.295	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 2.7\text{k}}}$	0.8033	-0.638
total	**0.237** (product)	$-$**3.78** (sum)

$$|H(j\omega_{in})| = 0.237, \qquad \angle H(j\omega_{in}) = -3.78 \text{ rad}$$
$$v_{o,ac}(t) = 0.237 \times 0.05\,\sin(2\pi \cdot 2000\,t - 3.78) = 0.0118\,\sin(2\pi \cdot 2000\,t - 3.78)$$

Add the DC part from Step 2 (superposition):

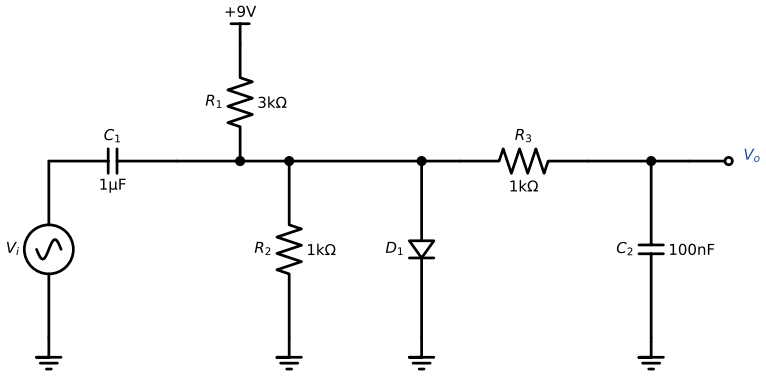
$$\boxed{V_o(t) = -0.95 + 0.0118\,\sin(2\pi \cdot 2000\,t - 3.78)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.81 mV, well below $nV_T = 45$ mV ✓.

Answer

$$H(s) = -0.295\,\frac{1}{1+\dfrac{s}{2\pi \cdot 2.7\text{k}}} \quad \text{(corners: 2.7 kHz)}$$
$$V_o(t) = -0.95 + 0.0118\,\sin(2\pi \cdot 2000\,t - 3.78)\ \text{V}$$

Solution 3D.12 **Medium**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). Assume the diode is ON (constant-voltage-drop model, 0.7 V) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_1 - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_1: && \frac{V_1 - 9}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_o}{R_3} + I_{D_1} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_1}{R_3} = 0 \\ &\text{D1 ON}: && V_1 - 0 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.7$$
$$I_{D_1} = 2.067 \text{ mA}$$

$I_{D_1} = 2.07\,\text{mA} > 0$ ✓, so the assumption holds.

$$\boxed{V_{o,dc} = 700\,\text{mV}}$$

Step 2 — Small-signal resistance

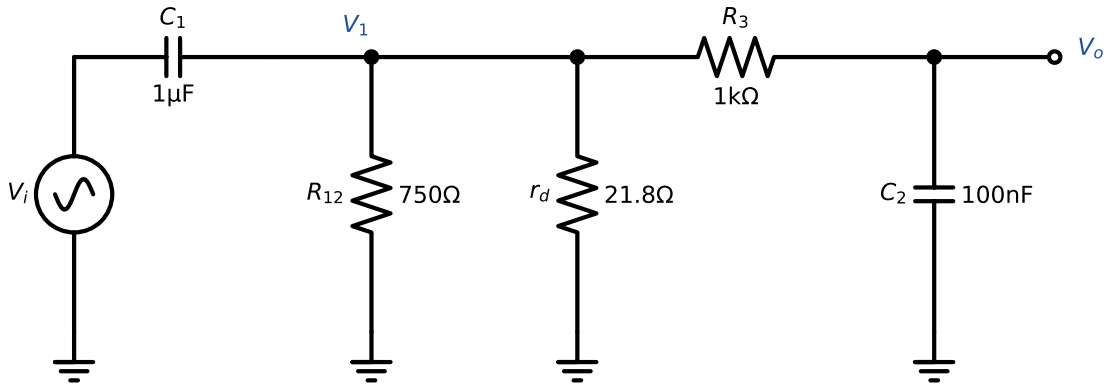
$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{2.07\,\text{mA}} = 21.8\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 3\,\text{k} \parallel 1\,\text{k} = 750\,\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel r_d \parallel \left(R_3 + \frac{1}{sC_2}\right) = \frac{20.7\,s + 2.07 \times 10^5}{s + 9790}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10000\,s}{s^2 + 58300\,s + 4.73 \times 10^8}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_3 + \frac{1}{sC_2}} = \frac{10000}{s + 10000}$$

Multiply the links and simplify:

$$H(s) = \frac{10000\,s}{s^2 + 58300\,s + 4.73 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 58300\,s + 4.73 \times 10^8$; quadratic formula:

$$s = \frac{-58300 \pm \sqrt{(58300)^2 - 4(4.73 \times 10^8)}}{2} = \frac{-58300 \pm 38800}{2} \Rightarrow s = -9740,\ -48500$$

Factored form (leading coefficient in front):

$$H(s) = 10000\,\frac{s}{(s+9740)(s+48500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.206\,\frac{s}{s+2\pi\cdot 1.55\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 7.72\text{k}}}, \qquad K = 0.206$$

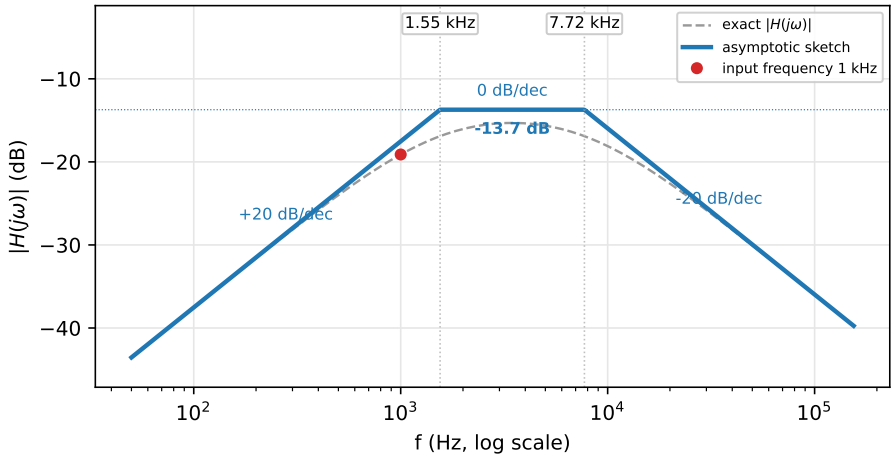
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	9740	1.55 kHz	-20 dB/dec
pole	48500	7.72 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 2.12\times10^{-5}\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.55$ kHz	+20 dB/dec	-13.7 dB at end
1.55 kHz – 7.72 kHz	0 dB/dec (flat)	-13.7 dB at start, -13.7 dB at end
$f > 7.72$ kHz	-20 dB/dec	-13.7 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.206 = -13.7$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.206	0.2061	0
$\dfrac{s}{s+2\pi\cdot 1.55\text{k}}$	0.5421	0.998
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 7.72\text{k}}}$	0.9917	−0.129
total	**0.1108** (product)	**0.869** (sum)

$$|H(j\omega_{in})| = 0.111, \qquad \angle H(j\omega_{in}) = 0.869 \text{ rad}$$

$$v_{o,ac}(t) = 0.111 \times 0.002\,\sin(2\pi\cdot 1000\,t + 0.869) = 2.22\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.869)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 0.7 + 2.22\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.869)\ \text{V}}$$

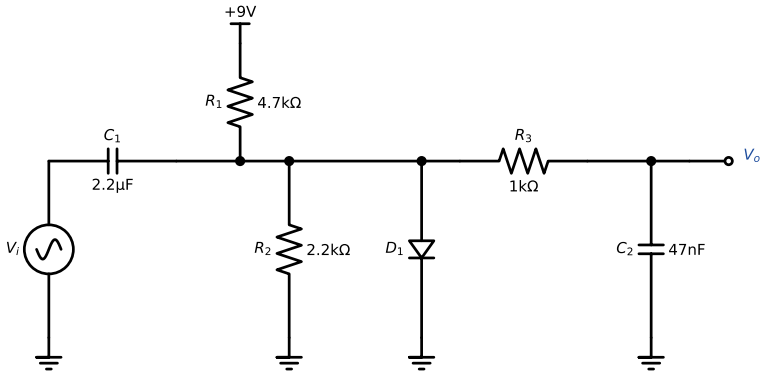
Check the small-signal assumption: the AC voltage across the diode has amplitude about $262\,\mu$V, well below $nV_T = 45\,$mV ✓.

Answer

$$H(s) = 0.206\,\frac{s}{s+2\pi\cdot 1.55\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 7.72\text{k}}} \quad \text{(corners: 1.55 kHz, 7.72 kHz)}$$

$$V_o(t) = 0.7 + 2.22\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.869)\ \text{V}$$

Solution 3D.13 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). With the exponential model the diode voltage is unknown, so first reduce the rest of the circuit to what the diode sees: a Thevenin source. Remove the diode and solve the circuit (Lecture 2 KCL): the open-circuit voltage across it is $V_{th} = 2.87\,\text{V}$. Short the diode instead: the current is $I_{sc} = 1.91\,\text{mA}$. So $R_{th} = \frac{V_{th}}{I_{sc}} = 1.5\,\text{k}\Omega$.

Two equations, two unknowns: the circuit gives $I_D = \frac{V_{th} - V_D}{R_{th}}$ and the diode gives $V_D = nV_T \ln \frac{I_D}{I_S}$ (tool T1, $n = 1$, $V_T = 25\,\text{mV}$, $I_S = 1\,\text{pA}$). Solve by iteration: guess $V_D = 0.7\,\text{V}$, compute I_D, update V_D, repeat until nothing changes:

step	V_D used (V)	circuit $\Rightarrow I_D$	diode $\Rightarrow$ new V_D (V)
1	0.7	$I_D = \frac{2.87-0.7}{1499} = 1.45\,\text{mA}$	0.5273
2	0.5273	$I_D = \frac{2.87-0.5273}{1499} = 1.56\,\text{mA}$	0.5292
3	0.5292	$I_D = \frac{2.87-0.5292}{1499} = 1.56\,\text{mA}$	0.5292
4	0.5292	$I_D = \frac{2.87-0.5292}{1499} = 1.56\,\text{mA}$	0.5292
5	0.5292	$I_D = \frac{2.87-0.5292}{1499} = 1.56\,\text{mA}$	0.5292

Converged: $V_D = 0.5292\,\text{V}$, $I_D = 1.56\,\text{mA}$. With this V_D the rest of the DC circuit gives $V_{o,dc} = 529\,\text{mV}$.

$$\boxed{V_{o,dc} = 529\,\text{mV}}$$

Step 2 — Small-signal resistance

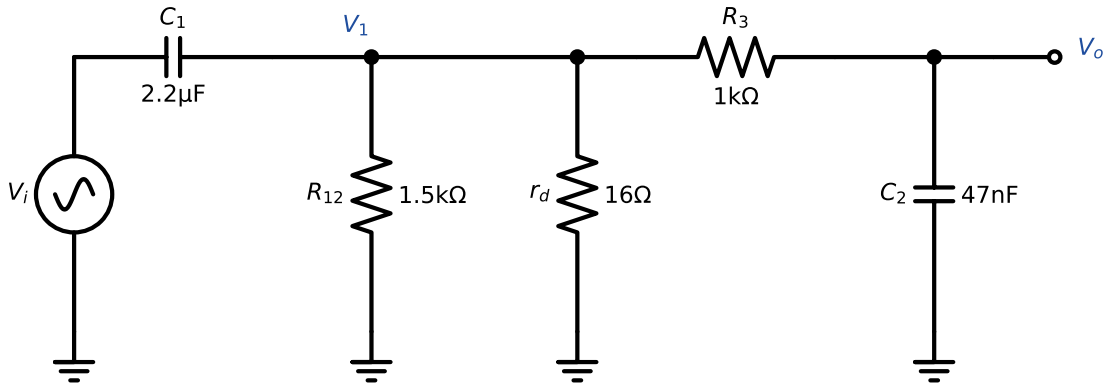
$r_d = \frac{nV_T}{I_D} = \frac{1 \times 25\,\text{mV}}{1.56\,\text{mA}} = 16\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 4.7\,\text{k} \parallel 2.2\,\text{k} = 1.5\,\text{k}\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel r_d \parallel \left(R_3 + \frac{1}{sC_2} \right) = \frac{15.6\, s + 3.32 \times 10^5}{s + 20900}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 21300\, s}{s^2 + 50400\, s + 6.11 \times 10^8}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_3 + \frac{1}{sC_2}} = \frac{21300}{s + 21300}$$

Multiply the links and simplify:

$$H(s) = \frac{21300\, s}{s^2 + 50400\, s + 6.11 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 50400\, s + 6.11 \times 10^8$; quadratic formula:

$$s = \frac{-50400 \pm \sqrt{(50400)^2 - 4(6.11 \times 10^8)}}{2} = \frac{-50400 \pm 10000}{2} \Rightarrow s = -20200,\ -30200$$

Factored form (leading coefficient in front):

$$H(s) = 21300\,\frac{s}{(s+20200)(s+30200)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.704\,\frac{s}{s+2\pi\cdot 3.21\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 4.81\text{k}}}, \qquad K = 0.704$$

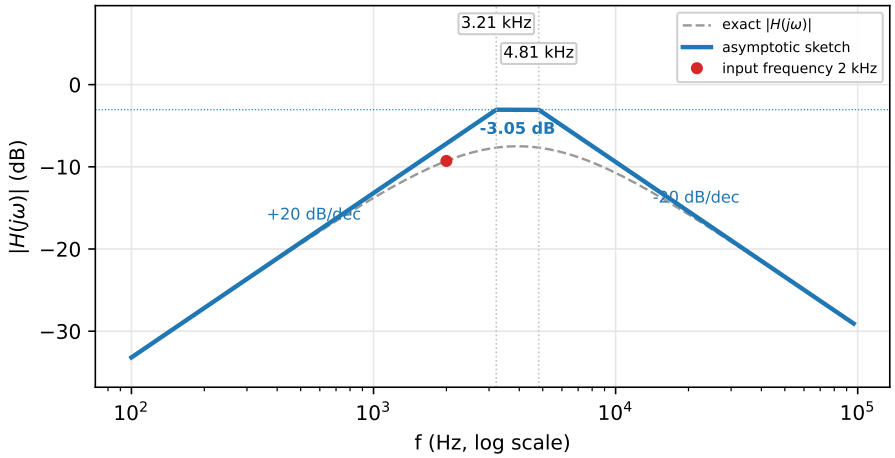
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	20200	3.21 kHz	-20 dB/dec
pole	30200	4.81 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 3.48\times10^{-5}\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.21$ kHz	$+20$ dB/dec	-3.05 dB at end
3.21 kHz – 4.81 kHz	0 dB/dec (flat)	-3.05 dB at start, -3.05 dB at end
$f > 4.81$ kHz	-20 dB/dec	-3.05 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.704 = -3.05$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.003\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.704	0.7038	0
$\dfrac{s}{s + 2\pi \cdot 3.21\text{k}}$	0.5283	1.01
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 4.81\text{k}}}$	0.9234	−0.394
total	**0.3433** (product)	**0.62** (sum)

$$|H(j\omega_{in})| = 0.343, \qquad \angle H(j\omega_{in}) = 0.62 \text{ rad}$$
$$v_{o,ac}(t) = 0.343 \times 0.003 \sin(2\pi \cdot 2000\,t + 0.62) = 0.00103 \sin(2\pi \cdot 2000\,t + 0.62)$$

Add the DC part from Step 2 (superposition):

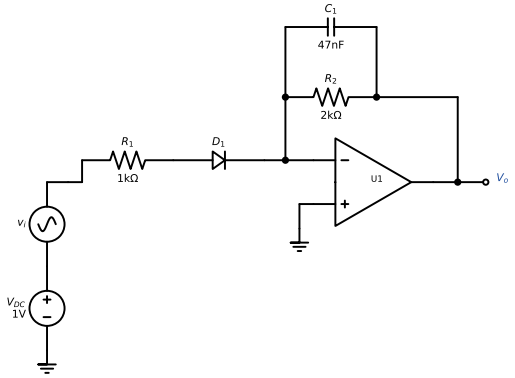
$$\boxed{V_o(t) = 0.529 + 0.00103 \sin(2\pi \cdot 2000\,t + 0.62)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 1.2 mV, well below $nV_T = 25\,\text{mV}$ ✓.

Answer

$$H(s) = 0.704\,\frac{s}{s + 2\pi \cdot 3.21\text{k}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 4.81\text{k}}} \quad \text{(corners: 3.21 kHz, 4.81 kHz)}$$
$$V_o(t) = 0.529 + 0.00103 \sin(2\pi \cdot 2000\,t + 0.62)\ \text{V}$$

Solution 3D.14 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 1\,\text{V}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The diode and R_1 connect V_{DC} to the virtual ground: the diode sees $V_{th} = V_{DC}$ through $R_{th} = R_1 = 1\,\text{k}\Omega$.

Two equations, two unknowns: the circuit gives $I_D = \frac{V_{th} - V_D}{R_{th}}$ and the diode gives $V_D = nV_T \ln \frac{I_D}{I_S}$ (tool T1, $n = 1.8$, $V_T = 25\,\text{mV}$, $I_S = 2.6\,\text{nA}$). Solve by iteration: guess $V_D = 0.7\,\text{V}$, compute I_D, update V_D, repeat until nothing changes:

step	V_D used (V)	circuit $\Rightarrow I_D$	diode $\Rightarrow$ new V_D (V)
1	0.7	$I_D = \frac{1-0.7}{1000} = 300\,\mu\text{A}$	0.5245
2	0.5245	$I_D = \frac{1-0.5245}{1000} = 475\,\mu\text{A}$	0.5452
3	0.5452	$I_D = \frac{1-0.5452}{1000} = 455\,\mu\text{A}$	0.5432
4	0.5432	$I_D = \frac{1-0.5432}{1000} = 457\,\mu\text{A}$	0.5434
5	0.5434	$I_D = \frac{1-0.5434}{1000} = 457\,\mu\text{A}$	0.5434

Converged: $I_D = 457\,\mu\text{A}$. All of I_D continues through the feedback resistor R_2 (the capacitor is open), so

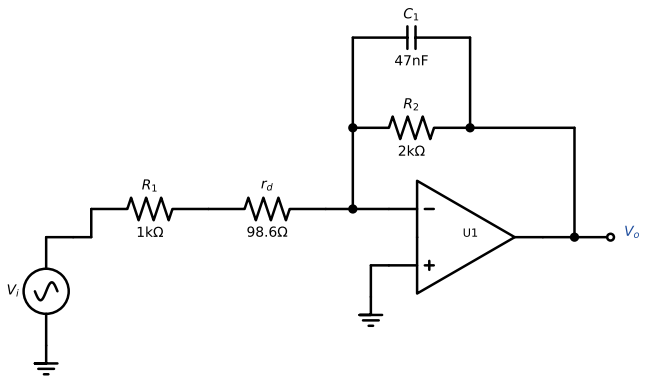
$$V_{o,dc} = 0 - R_2 I_D = -2000 \times 4.566 \times 10^{-4} = -913\,\text{mV}$$

$$\boxed{V_{o,dc} = -913\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{457\,\mu\text{A}} = 98.6\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + r_d = 1100$$

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{2.13 \times 10^7}{s + 10600}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-19400}{s + 10600}$$

So:

$$H(s) = \frac{-19400}{s + 10600}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10600$; $s + 10600 = 0 \Rightarrow s = -10600$.

Factored form (leading coefficient in front):

$$H(s) = -19400 \, \frac{1}{(s + 10600)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1.82 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.69\text{k}}}, \qquad K = -1.82$$

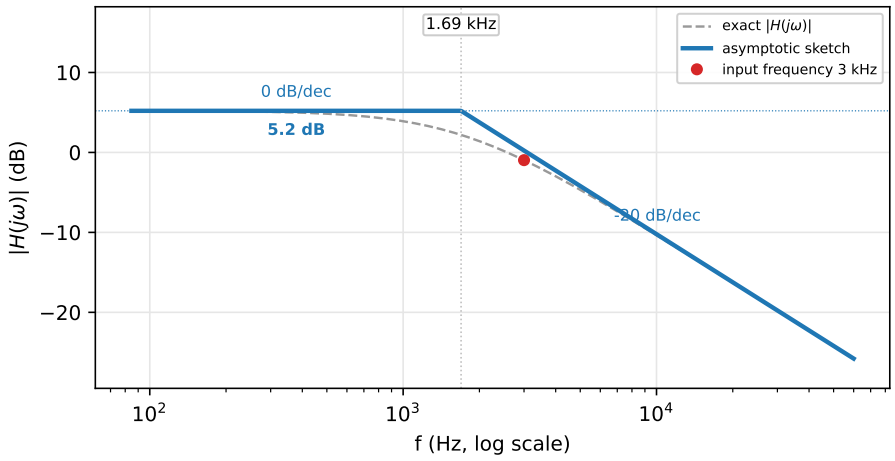
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10600	1.69 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1.82 = 5.2\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.69$ kHz	0 dB/dec (flat)	5.2 dB at end
$f > 1.69$ kHz	-20 dB/dec	5.2 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1.82 = 5.2\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.01\sin\left(2\pi\cdot 3000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi\cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.82	1.821	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.69\text{k}}}$	0.4915	-1.06
total	**0.8948** (product)	$\mathbf{-4.2}$ (sum)

$$|H(j\omega_{in})| = 0.895, \qquad \angle H(j\omega_{in}) = -4.2\ \text{rad}$$
$$v_{o,ac}(t) = 0.895\times 0.01\,\sin(2\pi\cdot 3000\,t - 3.41) = 0.00895\,\sin(2\pi\cdot 3000\,t - 3.41)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-4.2) = -3.41\,\text{rad}$.) Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = -0.913 + 0.00895\,\sin(2\pi\cdot 3000\,t - 3.41)\ \text{V}}$$

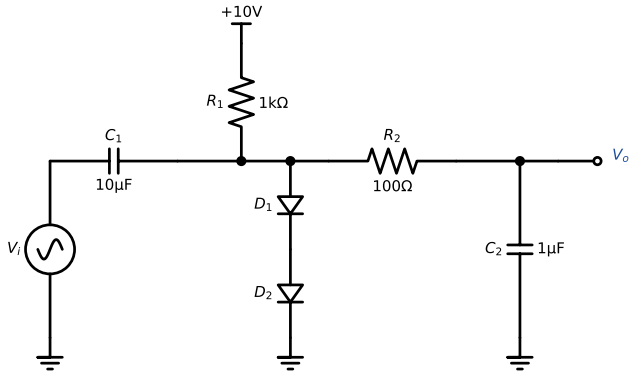
Check the small-signal assumption: the AC voltage across the diode has amplitude about $897\,\mu\text{V}$, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = -1.82\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.69\text{k}}} \quad \text{(corners: 1.69 kHz)}$$

$$V_o(t) = -0.913 + 0.00895\,\sin\left(2\pi \cdot 3000\,t - 3.41\right)\ \text{V}$$

Solution 3D.15 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). Assume both diodes are ON (0.7 V each) and solve the DC circuit:

Replace D_1 ON: a 0.7 V source ($V_1 - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 ON: a 0.7 V source ($V_a - 0 = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_1: && \frac{V_1 - 10}{R_1} + \frac{V_1 - V_o}{R_2} + I_{D_1} = 0 \\
&\text{KCL at } V_a: && -I_{D_1} + I_{D_2} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_2} = 0 \\
&\text{D1 ON}: && V_1 - V_a = 0.7 \\
&\text{D2 ON}: && V_a - 0 = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 1.4 \\
I_{D_1} &= 8.6 \text{ mA} \\
I_{D_2} &= 8.6 \text{ mA}
\end{aligned}$$

$I_{D_1} = 8.6\,\text{mA} > 0$, $I_{D_2} = 8.6\,\text{mA} > 0$ ✓, so the assumption holds.

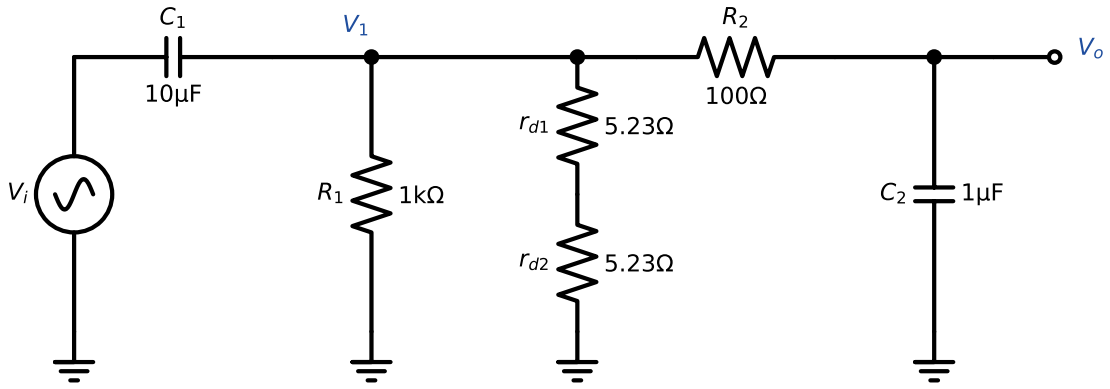
$$\boxed{V_{o,dc} = 1.4\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{8.6\,\text{mA}} = 5.23\,\Omega$ $r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{8.6\,\text{mA}} = 5.23\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel (r_{d1} + r_{d2}) \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{9.38\,s + 93800}{s + 9060}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10000\,s}{s^2 + 20700\,s + 9.66 \times 10^7}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{10000}{s + 10000}$$

Multiply the links and simplify:

$$H(s) = \frac{10000\,s}{s^2 + 20700\,s + 9.66 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 20700\,s + 9.66 \times 10^7$; quadratic formula:

$$s = \frac{-20700 \pm \sqrt{(20700)^2 - 4(9.66 \times 10^7)}}{2} = \frac{-20700 \pm 6360}{2} \Rightarrow s = -7150,\ -13500$$

Factored form (leading coefficient in front):

$$H(s) = 10000\,\frac{s}{(s+7150)(s+13500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.74\,\frac{s}{s+2\pi\cdot 1.14\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2.15\text{k}}}, \qquad K = 0.74$$

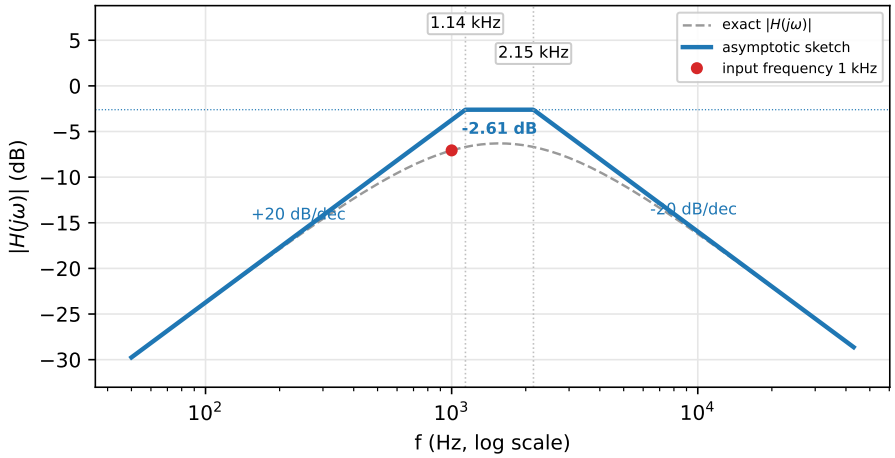
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	7150	1.14 kHz	−20 dB/dec
pole	13500	2.15 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.04 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 1.14 kHz	+20 dB/dec	-2.61 dB at end
1.14 kHz – 2.15 kHz	0 dB/dec (flat)	-2.61 dB at start, -2.61 dB at end
$f >$ 2.15 kHz	−20 dB/dec	-2.61 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.74 = -2.61$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.002\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.74	0.7404	0
$\dfrac{s}{s+2\pi\cdot 1.14\text{k}}$	0.6602	0.85
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2.15\text{k}}}$	0.9067	−0.435
total	**0.4432** (product)	**0.414** (sum)

$$|H(j\omega_{in})| = 0.443, \qquad \angle H(j\omega_{in}) = 0.414 \text{ rad}$$

$$v_{o,ac}(t) = 0.443 \times 0.002\,\sin(2\pi\cdot 1000\,t + 0.414) = 8.86\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.414)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 1.4 + 8.86\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.414)\ \text{V}}$$

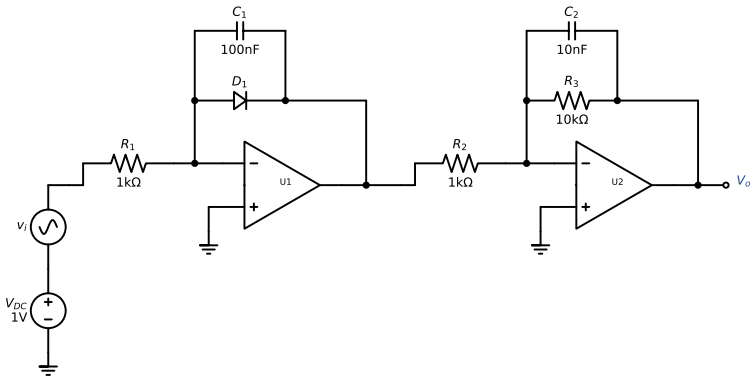
Check the small-signal assumption: the AC voltage across the diode has amplitude about $523\,\mu$V, well below $nV_T = 45\,$mV ✓.

Answer

$$H(s) = 0.74\,\frac{s}{s+2\pi\cdot 1.14\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2.15\text{k}}} \quad \text{(corners: 1.14 kHz, 2.15 kHz)}$$

$$V_o(t) = 1.4 + 8.86\times 10^{-4}\,\sin(2\pi\cdot 1000\,t + 0.414)\ \text{V}$$

Solution 3D.16 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 1\,\text{V}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The input current $\frac{V_{DC}-0}{R_1} = 1\,\text{mA}$ cannot enter the op-amp: all of it flows through the diode, $I_D = 1\,\text{mA}$. Exponential model (tool T1):

$$V_D = nV_T \ln \frac{I_D}{I_S} = 1.8 \times 0.025 \times \ln \frac{0.001}{2.6 \times 10^{-9}} = 579\,\text{mV}$$

The diode current flows from the − input (at 0 V) toward the output, so the output sits V_D below it:

$$V_{o,dc} = -(V_D) = -579\,\text{mV}$$

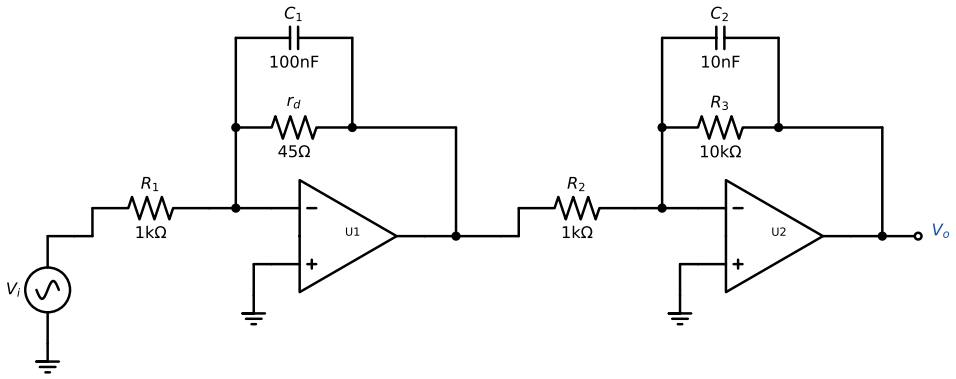
Block 2 passes DC with its gain at $s = 0$, $H_2(0) = -10$: $V_{dc} = -10 \times -0.5787 = 5.79\,\text{V}$.

$$\boxed{V_{o,dc} = 5.79\,\text{V}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8\times 25\,\text{mV}}{1\,\text{mA}} = 45\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_{o1} = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = r_d \parallel \frac{1}{sC_1} = \frac{10^7}{s + 2.22 \times 10^5}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10000}{s + 2.22 \times 10^5}$$

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{R_2}$ through R_2 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_2}$.

$$Z_{f2} = R_3 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 10000}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_2} = \frac{-10^5}{s + 10000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{10^9}{s^2 + 2.32 \times 10^5\, s + 2.22 \times 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 2.32 \times 10^5\, s + 2.22 \times 10^9$; quadratic formula:

$$s = \frac{-2.32 \times 10^5 \pm \sqrt{(2.32 \times 10^5)^2 - 4(2.22 \times 10^9)}}{2} = \frac{-2.32 \times 10^5 \pm 2.12 \times 10^5}{2} \Rightarrow s = -10000,\ -2.22{\times}10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^9 \frac{1}{(s + 10000)(s + 2.22 \times 10^5)}$$

Bode-ready form: rewrite every $(s + p) = p\left(1 + \frac{s}{p}\right)$ and $(s + z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.45 \frac{1}{1 + \frac{s}{2\pi \cdot 1.59\text{k}}} \frac{1}{1 + \frac{s}{2\pi \cdot 35.4\text{k}}}, \qquad K = 0.45$$

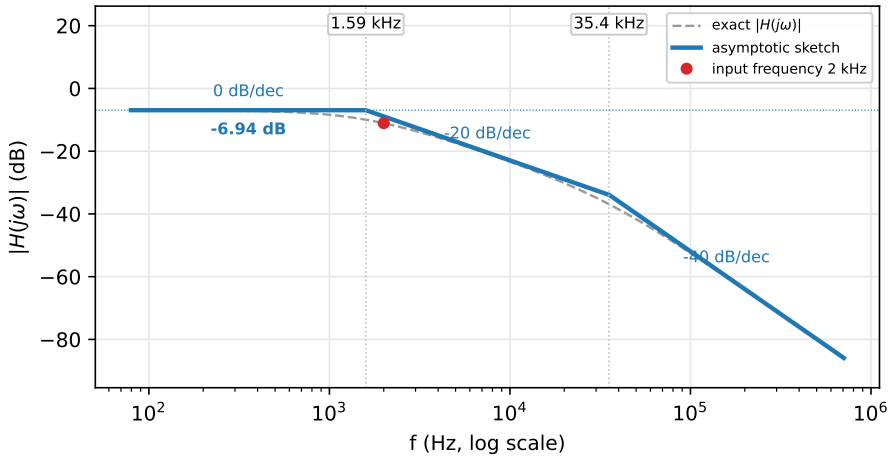
type	ω_c (rad/s)	$f_c = \omega_c / 2\pi$	slope change
pole	10000	1.59 kHz	−20 dB/dec
pole	2.22×10^5	35.4 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.45 = -6.94\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	-6.94 dB at end
1.59 kHz – 35.4 kHz	-20 dB/dec	-6.94 dB at start, -33.9 dB at end
$f > 35.4$ kHz	-40 dB/dec	-33.9 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.45 = -6.94\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.02\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.45	0.45	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.6227	-0.899
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 35.4\text{k}}}$	0.9984	-0.0565
total	**0.2798** (product)	$\mathbf{-0.955}$ (sum)

$$|H(j\omega_{in})| = 0.28, \qquad \angle H(j\omega_{in}) = -0.955 \text{ rad}$$
$$v_{o,ac}(t) = 0.28 \times 0.02 \sin(2\pi \cdot 2000\,t - 0.955) = 0.0056 \sin(2\pi \cdot 2000\,t - 0.955)$$

Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 5.79 + 0.0056 \sin(2\pi \cdot 2000\,t - 0.955)\ \text{V}}$$

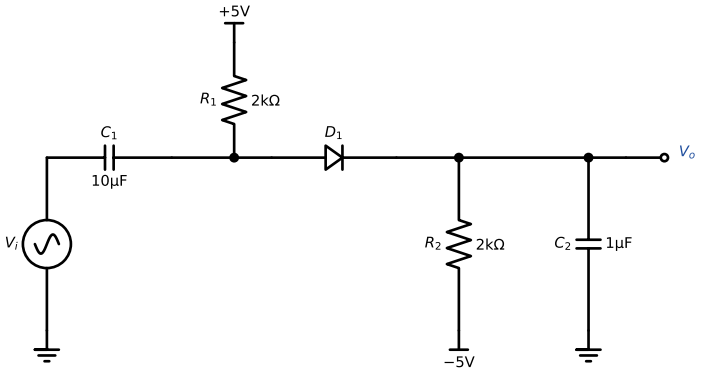
Check the small-signal assumption: the AC voltage across the diode has amplitude about $899\,\mu$V, well below $nV_T = 45\,\text{mV}$ ✓.

Answer

$$H(s) = 0.45\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 35.4\text{k}}} \quad \text{(corners: 1.59 kHz, 35.4 kHz)}$$

$$V_o(t) = 5.79 + 0.0056\,\sin\left(2\pi\cdot 2000\,t - 0.955\right)\ \text{V}$$

Solution 3D.17 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$ and every capacitor is open (C_1, C_2). With the exponential model the diode voltage is unknown, so first reduce the rest of the circuit to what the diode sees: a Thevenin source. Remove the diode and solve the circuit (Lecture 2 KCL): the open-circuit voltage across it is $V_{th} = 10\,\text{V}$. Short the diode instead: the current is $I_{sc} = 2.5\,\text{mA}$. So $R_{th} = \frac{V_{th}}{I_{sc}} = 4\,\text{k}\Omega$.

Two equations, two unknowns: the circuit gives $I_D = \frac{V_{th} - V_D}{R_{th}}$ and the diode gives $V_D = nV_T \ln \frac{I_D}{I_S}$ (tool T1, $n = 1.8$, $V_T = 25\,\text{mV}$, $I_S = 2.6\,\text{nA}$). Solve by iteration: guess $V_D = 0.7\,\text{V}$, compute I_D, update V_D, repeat until nothing changes:

step	V_D used (V)	circuit $\Rightarrow I_D$	diode $\Rightarrow$ new V_D (V)
1	0.7	$I_D = \frac{10-0.7}{4000} = 2.33\,\text{mA}$	0.6167
2	0.6167	$I_D = \frac{10-0.6167}{4000} = 2.35\,\text{mA}$	0.6171
3	0.6171	$I_D = \frac{10-0.6171}{4000} = 2.35\,\text{mA}$	0.6171
4	0.6171	$I_D = \frac{10-0.6171}{4000} = 2.35\,\text{mA}$	0.6171
5	0.6171	$I_D = \frac{10-0.6171}{4000} = 2.35\,\text{mA}$	0.6171

Converged: $V_D = 0.6171\,\text{V}$, $I_D = 2.35\,\text{mA}$. With this V_D the rest of the DC circuit gives $V_{o,dc} = -309\,\text{mV}$.

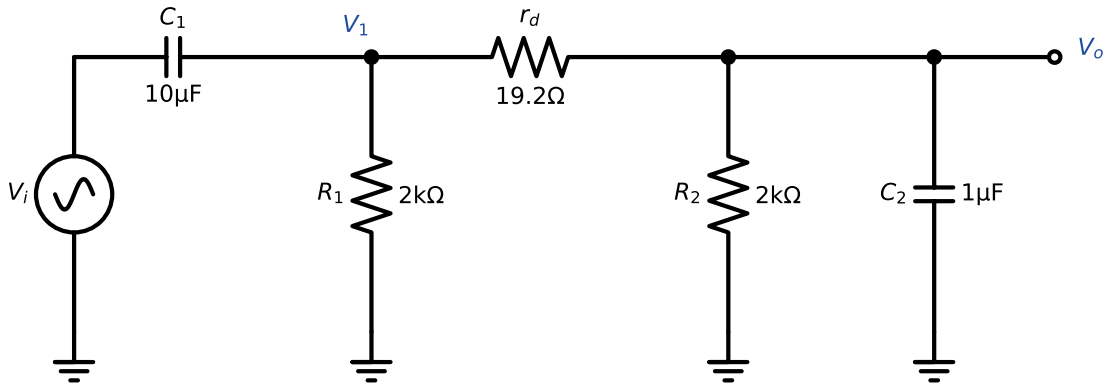
$$\boxed{V_{o,dc} = -309\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1.8 \times 25\,\text{mV}}{2.35\,\text{mA}} = 19.2\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor.

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \frac{1}{sC_2} = \frac{10^6}{s + 500}$$
$$Z_1 = R_1 \parallel (r_d + Z_o) = \frac{19\,s + 10^6}{s + 995}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 52600\,s}{s^2 + 57900\,s + 5.24 \times 10^6}$$
$$\frac{V_o}{V_1} = \frac{Z_o}{r_d + Z_o} = \frac{52100}{s + 52600}$$

Multiply the links and simplify:

$$H(s) = \frac{52100\,s}{s^2 + 57900\,s + 5.24 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 57900\,s + 5.24 \times 10^6$; quadratic formula:

$$s = \frac{-57900 \pm \sqrt{(57900)^2 - 4(5.24 \times 10^6)}}{2} = \frac{-57900 \pm 57700}{2} \Rightarrow s = -90.6,\ -57800$$

Factored form (leading coefficient in front):

$$H(s) = 52100\,\frac{s}{(s+90.6)(s+57800)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.902\,\frac{s}{s+2\pi\cdot 14.4}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 9.2\text{k}}}, \qquad K = 0.902$$

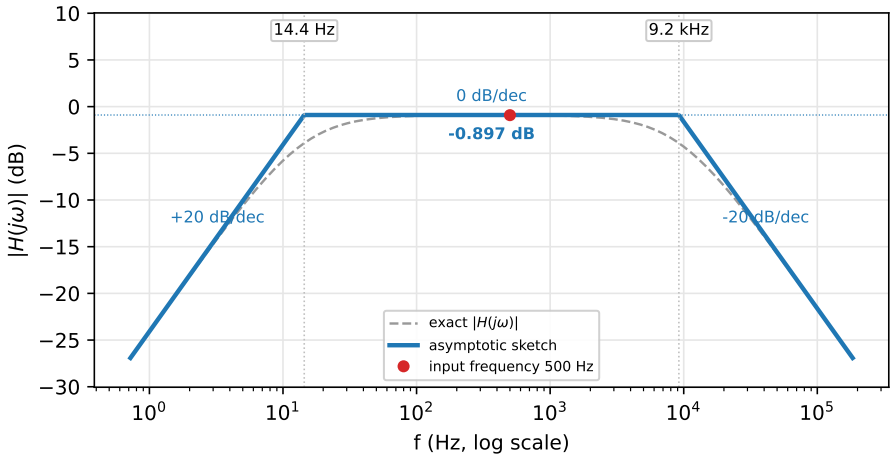
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	90.6	14.4 Hz	−20 dB/dec
pole	57800	9.2 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00995\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 14.4 Hz	+20 dB/dec	-0.897 dB at end
14.4 Hz – 9.2 kHz	0 dB/dec (flat)	-0.897 dB at start, -0.897 dB at end
$f >$ 9.2 kHz	−20 dB/dec	-0.897 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.902 = -0.897$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.005\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.902	0.9019	0
$\dfrac{s}{s+2\pi\cdot 14.4}$	0.9996	0.0288
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 9.2\text{k}}}$	0.9985	−0.0543
total	**0.9002** (product)	**−0.0255** (sum)

$$|H(j\omega_{in})| = 0.9, \qquad \angle H(j\omega_{in}) = -0.0255 \text{ rad}$$
$$v_{o,ac}(t) = 0.9 \times 0.005\,\sin(2\pi\cdot 500\,t - 0.0255) = 0.0045\,\sin(2\pi\cdot 500\,t - 0.0255)$$

Add the DC part from Step 2 (superposition):

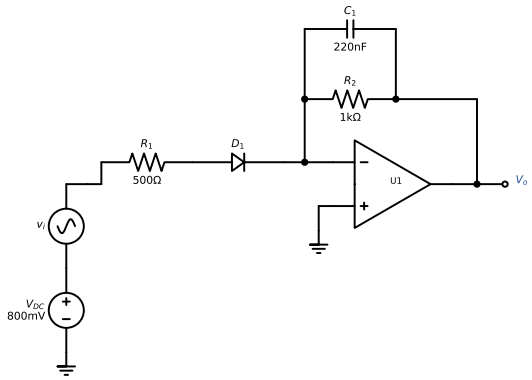
$$\boxed{V_o(t) = -0.309 + 0.0045\,\sin(2\pi\cdot 500\,t - 0.0255)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about $275\,\mu$V, well below $nV_T = 45\,$mV ✓.

Answer

$$H(s) = 0.902\,\frac{s}{s+2\pi\cdot 14.4}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 9.2\text{k}}} \quad \text{(corners: 14.4 Hz, 9.2 kHz)}$$
$$V_o(t) = -0.309 + 0.0045\,\sin(2\pi\cdot 500\,t - 0.0255)\ \text{V}$$

Solution 3D.18 **Hard**



Step 1 — DC operating point

DC rules (tool T3): $v_i = 0$, so the input is just $V_{DC} = 800\,\text{mV}$; capacitors are open. The op-amp has negative feedback, so $V_- = V_+ = 0$ (virtual ground) and no current enters its inputs. The diode and R_1 connect V_{DC} to the virtual ground: the diode sees $V_{th} = V_{DC}$ through $R_{th} = R_1 = 500\,\Omega$.

Two equations, two unknowns: the circuit gives $I_D = \frac{V_{th} - V_D}{R_{th}}$ and the diode gives $V_D = nV_T \ln \frac{I_D}{I_S}$ (tool T1, $n = 1$, $V_T = 25\,\text{mV}$, $I_S = 1\,\text{pA}$). Solve by iteration: guess $V_D = 0.7\,\text{V}$, compute I_D, update V_D, repeat until nothing changes:

step	V_D used (V)	circuit $\Rightarrow I_D$	diode $\Rightarrow$ new V_D (V)
1	0.7	$I_D = \frac{0.8-0.7}{500} = 200\,\mu\text{A}$	0.4778
2	0.4778	$I_D = \frac{0.8-0.4778}{500} = 644\,\mu\text{A}$	0.5071
3	0.5071	$I_D = \frac{0.8-0.5071}{500} = 586\,\mu\text{A}$	0.5047
4	0.5047	$I_D = \frac{0.8-0.5047}{500} = 591\,\mu\text{A}$	0.5049
5	0.5049	$I_D = \frac{0.8-0.5049}{500} = 590\,\mu\text{A}$	0.5049

Converged: $I_D = 590\,\mu\text{A}$. All of I_D continues through the feedback resistor R_2 (the capacitor is open), so

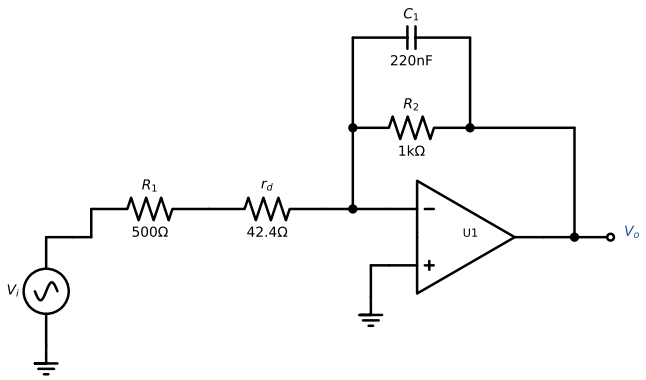
$$V_{o,dc} = 0 - R_2 I_D = -1000 \times 5.902 \times 10^{-4} = -590\,\text{mV}$$

$$\boxed{V_{o,dc} = -590\,\text{mV}}$$

Step 2 — Small-signal resistance

$r_d = \frac{nV_T}{I_D} = \frac{1 \times 25\,\text{mV}}{590\,\mu\text{A}} = 42.4\,\Omega$ (tool T3). For the AC analysis the diode is just this resistor, and V_{DC} is replaced by a short (AC ground).

Step 3 — AC circuit



Step 4 — Transfer function

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + r_d = 542$$

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{4.55 \times 10^6}{s + 4550}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-8380}{s + 4550}$$

So:

$$H(s) = \frac{-8380}{s + 4550}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 4550$; $s + 4550 = 0 \Rightarrow s = -4550$.

Factored form (leading coefficient in front):

$$H(s) = -8380 \, \frac{1}{(s + 4550)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1.84 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 723}}, \qquad K = -1.84$$

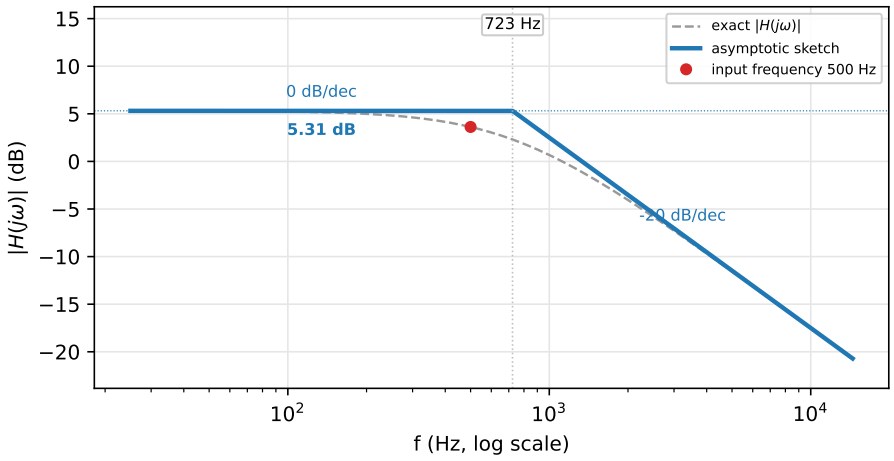
type	ω_c (rad/s)	$f_c = \omega_c / 2\pi$	slope change
pole	4550	723 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1.84 = 5.31\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 723$ Hz	0 dB/dec (flat)	5.31 dB at end
$f > 723$ Hz	-20 dB/dec	5.31 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1.84 = 5.31$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Total output (DC + AC)

The input is $V_i(t) = 0.005\cos\left(2\pi\cdot 500\,t - \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 500 = 3140$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.84	1.844	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 723}}$	0.8226	-0.605
total	**1.517** (product)	**-3.75** (sum)

$$|H(j\omega_{in})| = 1.52, \qquad \angle H(j\omega_{in}) = -3.75 \text{ rad}$$
$$v_{o,ac}(t) = 1.52\times 0.005\cos\left(2\pi\cdot 500\,t - 4.79\right) = 0.00758\cos\left(2\pi\cdot 500\,t - 4.79\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{3} + (-3.75) = -4.79$ rad.) Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = -0.59 + 0.00758\cos\left(2\pi\cdot 500\,t - 4.79\right)\ \text{V}}$$

Check the small-signal assumption: the AC voltage across the diode has amplitude about 391 μV, well below $nV_T = 25$ mV ✓.

Answer

$$H(s) = -1.84\,\frac{1}{1+\dfrac{s}{2\pi\cdot 723}} \quad \text{(corners: 723 Hz)}$$

$$V_o(t) = -0.59 + 0.00758\,\cos\left(2\pi\cdot 500\,t - 4.79\right)\ \text{V}$$